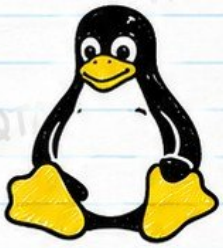
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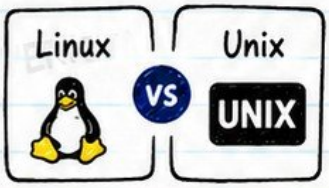
200 LINUX INTERVIEW Q&A

1. What is Linux?



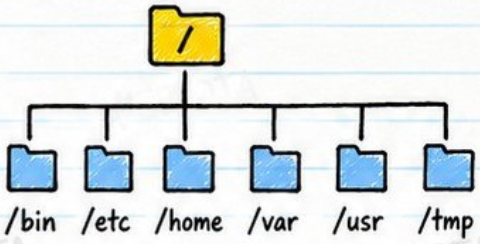
Answer: Linux is an open-source, Unix-like operating system kernel developed by Linus Torvalds. It is widely used in servers, desktops, mobile devices, and embedded systems.

2. What is the difference between Linux and Unix?



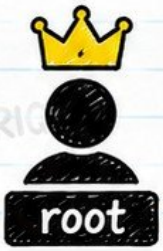
Answer: Linux is an open-source Unix-like operating system kernel, while Unix is a proprietary operating system developed by AT&T Bell Labs. Unix requires a license, whereas Linux is free and open-source.

3. Explain the Linux file system hierarchy.



Answer: The Linux file system hierarchy is structured in a tree-like format starting from the root directory (/). Important directories include /bin, /etc, /home, /var, /usr, and /tmp.

4. What is the root account?



Answer: The root account is the superuser account in Linux with unrestricted access to all commands, files, and system resources. It is used for system administration tasks.

5. How do you change file permissions in Linux?

- ✔ r (read) = 4
- ✔ w (write) = 2
- ✔ x (execute) = 1

Answer: Use the chmod command. For example, to set read, write, and execute permissions for the owner and read permissions for others, use:

```
[user@linux ~]$ chmod 744 filename
```

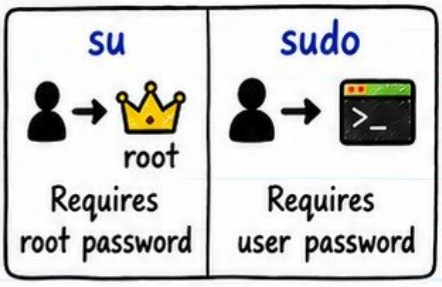
7 (rwx) Owner | 4 (r--) Group | 4 (r--) Others

Master the fundamentals. Think like a senior engineer. Solve real production problems.

200 LINUX INTERVIEW Q&A

6. What is the difference between su and sudo?

Answer: su switches the user to the root account, requiring the root password. sudo allows a permitted user to execute a command as the superuser or another user, requiring the user's password.



7. How do you check the current runlevel?

Answer: Use the runlevel command:

```
[user@linux ~]$ runlevel
```



8. How do you list all installed packages in a Debian-based system?

Answer: Use the dpkg command:

```
[user@linux ~]$ dpkg --list
```



9. How do you view running processes in Linux?

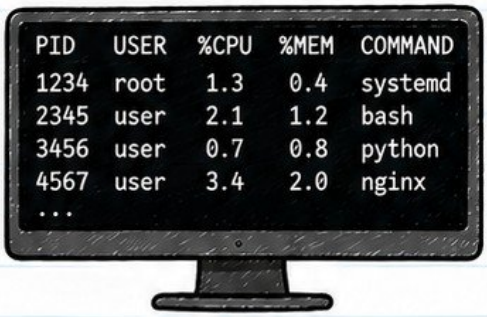
Answer: Use the ps or top command.

- For a snapshot:

```
[user@linux ~]$ ps aux
```

- For a dynamic view:

```
[user@linux ~]$ top
```



10. What is the use of the grep command?

Answer: grep searches for patterns within files. For example, to search for "error" in a file:

```
[user@linux ~]$ grep "error" filename
```



11. How do you display disk usage of directories?

Answer: Use the du command:

```
[user@linux ~]$ du -sh /path/to/directory
```



12. How do you check free disk space?

Answer: Use the df command:

```
[user@linux ~]$ df -h
```

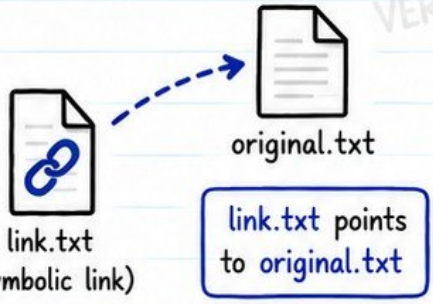
Filesystem	Size	Used	Avail	Use%	Mounted on
/dev/sda1	50G	20G	28G	42%	/
tmpfs	2.0G	0	2.0G	0%	/dev/shm
/dev/sdb1	100G	45G	50G	48%	/data

200 LINUX INTERVIEW Q&A

13. What is a symbolic link?

Answer: A symbolic link is a type of file that points to another file or directory. It is created using the ln -s command:

```
$ ln -s /path/to/target /path/to/link
```



14. How do you find files containing a specific string?

Answer: Use the grep command with -r for recursive search:

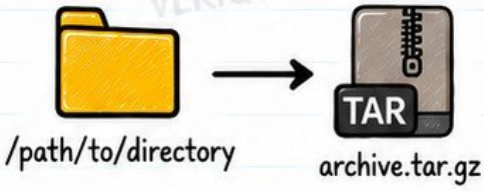
```
$ grep -r "search_string" /path/to/directory
```



15. How do you compress files using tar?

Answer: Use the tar command with -czf to create a compressed archive:

```
$ tar -czf archive.tar.gz /path/to/directory
```



16. How do you extract files from a tar archive?

Answer: Use the tar command with -xzf to extract:

```
$ tar -xzf archive.tar.gz
```



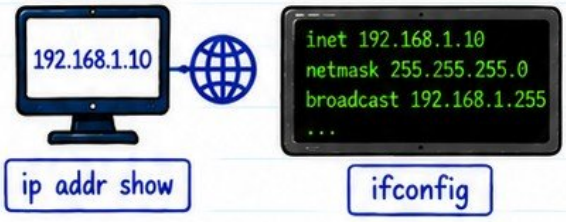
17. How do you find the IP address of your system?

Answer: Use the ip or ifconfig command:

```
$ ip addr show
```

or

```
$ ifconfig
```



18. How do you view the contents of a file?

Answer: Use commands like cat, less, or more. For example:

```
$ cat filename
```



19. How do you change the hostname of a Linux system?

Answer: Use the hostnamectl command:

```
$ sudo hostnamectl set-hostname newhostname
```



200 LINUX INTERVIEW Q&A

20. What is the crontab command used for?

Answer: crontab is used to schedule jobs, commands, or scripts, to run periodically.

For example, to edit the crontab file:

```
[user@linux ~]$ crontab -e
```



Crontab Format (Minute Hour Day Month DayOfWeek Command)

```
* * * * * /path/to/script.sh
```

Runs the script every minute

21. How do you add a user in Linux?

Answer: Use the adduser command:

```
[user@linux ~]$ sudo adduser username
```



22. How do you delete a user in Linux?

Answer: Use the deluser command:

```
[user@linux ~]$ sudo deluser username
```



23. How do you switch to another user account?

Answer: Use the su command:

```
[user@linux ~]$ su - username
```

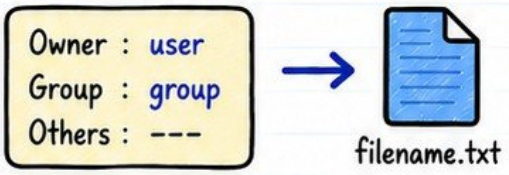
```
[user@linux ~]$ su - alice
Password:
[alice@linux ~]$
```

Switch to 'alice' user

24. How do you change file ownership?

Answer: Use the chown command:

```
[user@linux ~]$ sudo chown user:group filename
```



25. How do you view system logs?

Answer: Use the journalctl command for systemd logs or tail for log files. For example:

```
[user@linux ~]$ journalctl
```

or

```
[user@linux ~]$ tail -f /var/log/syslog
```

```
[user@linux ~]$ journalctl -n 5
May 26 10:15:22 linux systemd[1]: Started Network Service.
May 26 10:15:25 linux sshd[1234]: Accepted password for user.
May 26 10:15:30 linux CRON[2345]: (root) CMD (run-parts /etc/cron.hourly)
May 26 10:15:45 linux systemd[1]: Started Daily Cleanup.
May 26 10:16:01 linux kernel: [12345.678] EXT4-fs (sda1): mounted.
```

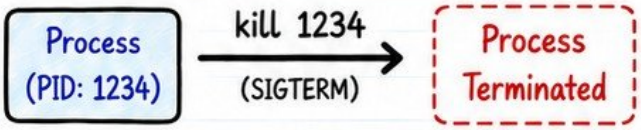


/var/log/syslog

26. How do you kill a process by its PID?

Answer: Use the kill command:

```
[user@linux ~]$ kill PID
```



200 LINUX INTERVIEW Q&A

27. How do you forcefully kill a process?

Answer: Use the kill -9 command:

```
[user@linux ~]$ kill -9 PID
```



28. How do you find the PID of a process by its name?

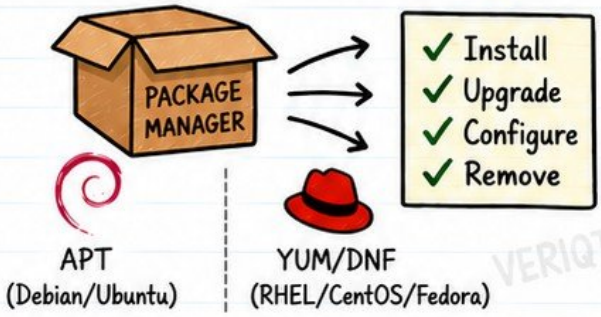
Answer: Use the pgrep command:

```
[user@linux ~]$ pgrep processname
```



29. What is a package manager?

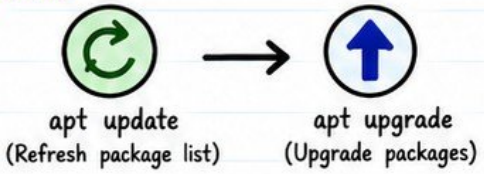
Answer: A package manager automates the process of installing, upgrading, configuring, and removing software packages. Examples include APT for Debian-based systems and YUM/DNF for Red Hat-based systems.



30. How do you update all packages in a Debian-based system?

Answer: Use the apt update and apt upgrade commands:

```
[user@linux ~]$ sudo apt update
[user@linux ~]$ sudo apt upgrade
```



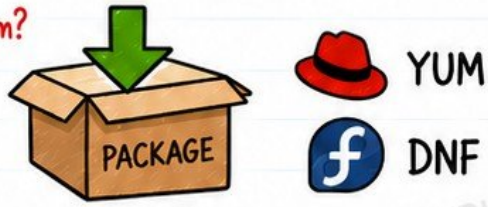
31. How do you install a package in a Red Hat-based system?

Answer: Use the yum or dnf command:

```
[user@linux ~]$ sudo yum install package_name
```

or

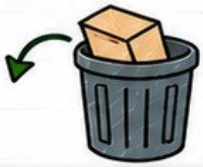
```
[user@linux ~]$ sudo dnf install package_name
```



32. How do you remove a package in a Debian-based system?

Answer: Use the apt remove command:

```
[user@linux ~]$ sudo apt remove package_name
```



33. What is the difference between apt-get and apt?

Answer: apt is a higher-level interface for package management, providing a more user-friendly set of commands that include the functionality of apt-get, apt-cache, and other lower-level tools.

apt-get	apt
Low-level tool	High-level tool
More options	Simpler & user-friendly
Part of older tools	Modern & recommended
Used in scripts	Used by default in most distributions

200 LINUX INTERVIEW Q&A

34. What is the find command used for?

Answer: find is used to search for files and directories based on various criteria. For example, to find all .txt files:

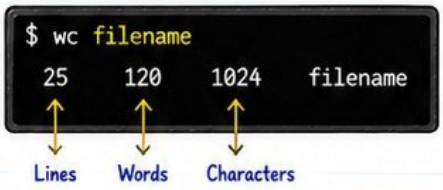
```
$ find /path/to/search -name "*.txt"
```



35. How do you count the number of lines, words, and characters in a file?

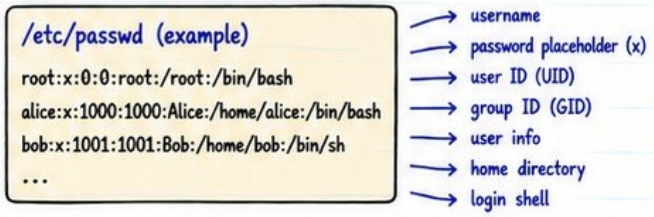
Answer: Use the wc command:

```
$ wc filename
```



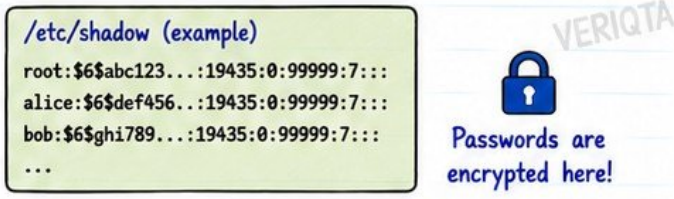
36. What is the purpose of the /etc/passwd file?

Answer: The /etc/passwd file contains user account information, including username, user ID, group ID, home directory, and shell.



37. What is the /etc/shadow file?

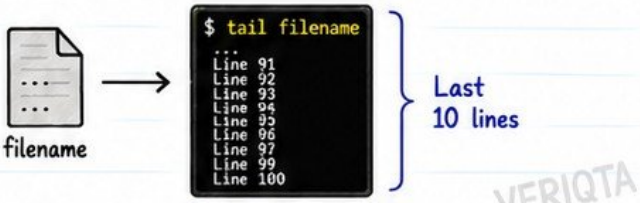
Answer: The /etc/shadow file contains encrypted user passwords and additional account information like password expiration.



38. How do you display the last 10 lines of a file?

Answer: Use the tail command:

```
$ tail filename
```



39. How do you display the first 10 lines of a file?

Answer: Use the head command:

```
$ head filename
```



40. How do you combine multiple files into one?

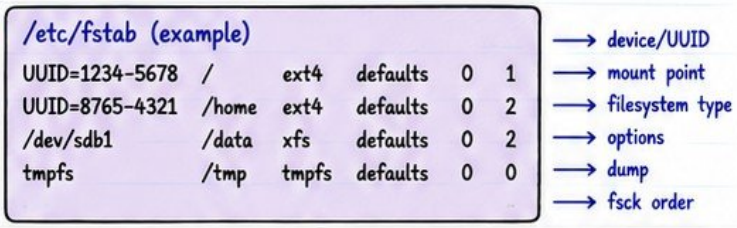
Answer: Use the cat command:

```
$ cat file1 file2 > combined_file
```



41. What is the purpose of the /etc/fstab file?

Answer: The /etc/fstab file contains information about file systems and their mount points, specifying how and where they should be mounted.

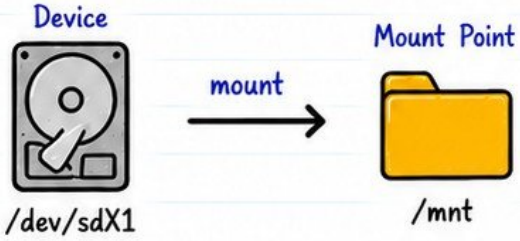


200 LINUX INTERVIEW Q&A

42. How do you mount a file system?

Answer: Use the mount command:

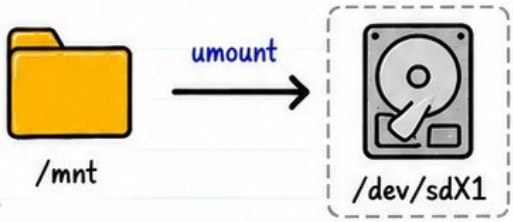
```
$ sudo mount /dev/sdX1 /mnt
```



43. How do you unmount a file system?

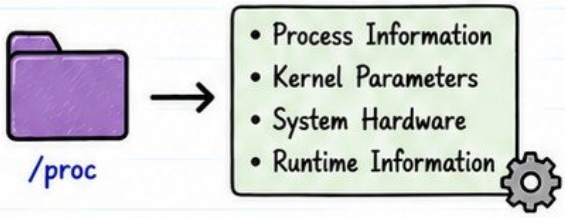
Answer: Use the umount command:

```
$ sudo umount /mnt
```



44. What is the purpose of the /proc directory?

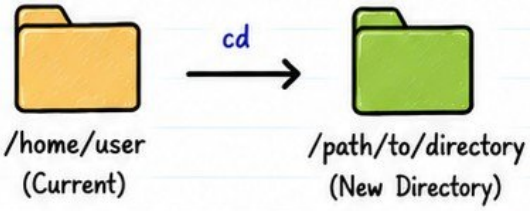
Answer: The /proc directory is a virtual file system providing information about system processes and kernel parameters.



45. How do you change the current directory?

Answer: Use the cd command:

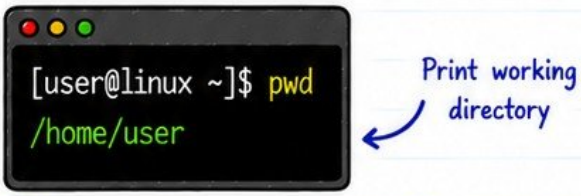
```
$ cd /path/to/directory
```



46. How do you display the current directory?

Answer: Use the pwd command:

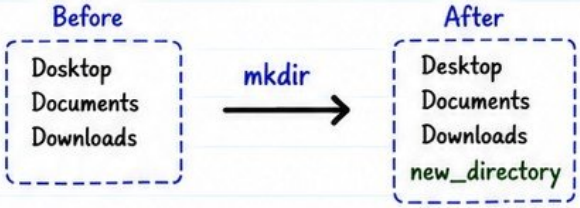
```
$ pwd
```



47. How do you create a new directory?

Answer: Use the mkdir command:

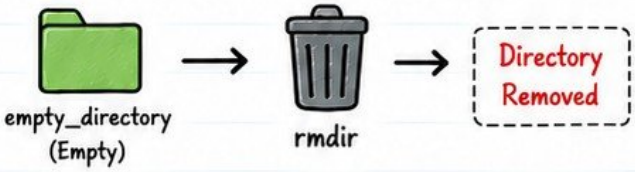
```
$ mkdir new_directory
```



48. How do you remove an empty directory?

Answer: Use the rmdir command:

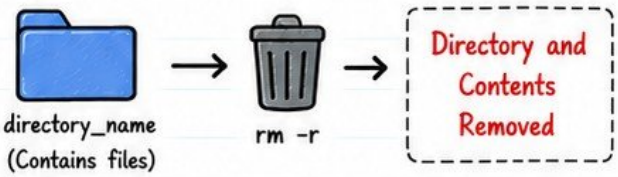
```
$ rmdir empty_directory
```



49. How do you remove a directory and its contents?

Answer: Use the rm -r command:

```
$ rm -r directory_name
```



200 LINUX INTERVIEW Q&A

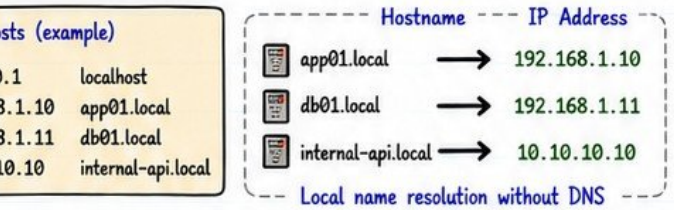
50. What is the purpose of the /etc/hosts file?

Answer: The /etc/hosts file maps hostnames to IP addresses, providing a way to associate names with IP addresses without using DNS.

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/etc/hosts (example)

127.0.0.1	localhost
192.168.1.10	app01.local
192.168.1.11	db01.local
10.10.10.10	internal-api.local



51. How do you display all open network connections?

Answer: Use the netstat command:

```
$ netstat -tuln
```

```
$ netstat -tuln
Proto  Recv-Q Send-Q Local Address     Foreign Address   State
tcp        0      0  0.0.0.0:22        0.0.0.0:*         LISTEN
tcp        0      0  127.0.0.1:25      0.0.0.0:*         LISTEN
udp        0      0  0.0.0.0:53        0.0.0.0:*
udp        0      0  127.0.0.1:123     0.0.0.0:*
```

Flags:
-t TCP
-u UDP
-l Listening
-n Numeric

52. What is the ifconfig command used for?

Answer: ifconfig is used to configure network interfaces, assign IP addresses, and manage network connections. However, ip is recommended for newer systems.

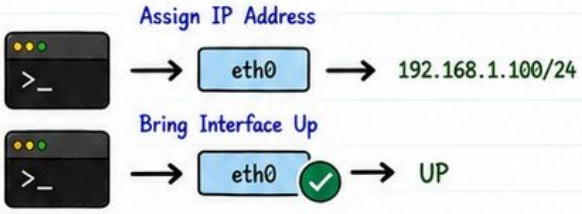
```
$ ifconfig eth0
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
      inet 192.168.1.100  netmask 255.255.255.0  broadcast 192.168.1.255
      inet6 fe80::a00:27ff:fe4e:66a1  prefixlen 64  scopeid 0x20<link>
      RX packets 12345  bytes 987654 (964.5 KiB)
      TX packets 6789  bytes 123456 (120.5 KiB)
```

TIP: Use 'ip' command for modern Linux distributions.

53. How do you configure a network interface using the ip command?

Answer: Use the ip addr command:

```
$ ip addr add 192.168.1.100/24 dev eth0
$ ip link set eth0 up
```



54. What is the ping command used for?

Answer: ping is used to check the network connectivity between two nodes by sending ICMP echo requests and measuring the response time.

```
$ ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=115 time=12.3 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=115 time=11.8 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=115 time=12.1 ms
--- 8.8.8.8 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss,
rtt min/avg/max/mdev = 11.8/12.0/12.3/0.2 ms
```



55. How do you check DNS resolution for a domain?

Answer: Use the dig or nslookup command:

```
$ dig example.com
or
$ nslookup example.com
```

```
dig example.com
;; ANSWER SECTION:
example.com.   300   IN   A   93.184.216.34
;; Query time: 25 msec
;; SERVER: 8.8.8.8#53(8.8.8.8)
;; WHEN: Tue May 21 10:20:00 UTC 2024
;; MSG SIZE  rcvd: 56
```

OR

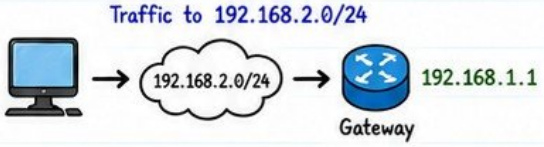
```
nslookup example.com
Server:    8.8.8.8
Address:   8.8.8.8#53

Name:      example.com
Address:   93.184.216.34
```

56. How do you add a new route in the routing table?

Answer: Use the ip route add command:

```
$ sudo ip route add 192.168.2.0/24 via 192.168.1.1
```



Use sudo to add routes permanently (may require persistence config).

57. How do you display the routing table?

Answer: Use the ip route show command:

```
$ ip route show
```

```
$ ip route show
default via 192.168.1.1 dev eth0
192.168.1.0/24 dev eth0 proto kernel scope link src 192.168.1.100
192.168.2.0/24 via 192.168.1.1 dev eth0
```

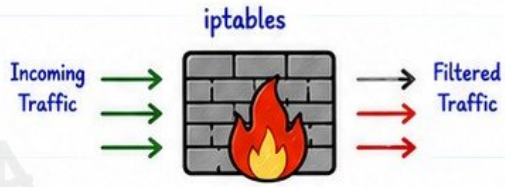
Key Columns:
- Destination
- Gateway
- Interface
- Source

PRO TIP (SENIOR ENGINEER INSIGHT): Know your system deeply. These commands form the foundation of production troubleshooting, performance analysis, incident response, and infrastructure automation.

VERIQTA
200 LINUX INTERVIEW Q&A

58. What is the iptables command used for?

Answer: iptables is used to set up, maintain, and inspect the tables of IP packet filter rules in the Linux kernel.



PRO TIP
iptables is stateful. Use conntrack to track connection states.

59. How do you list the current iptables rules?

Answer: Use the iptables -L command:

```
$ sudo iptables -L
```

```
$ sudo iptables -L
Chain INPUT (policy ACCEPT)
target     prot opt source               destination
ACCEPT     tcp  --  anywhere             anywhere             tcp dpt:ssh
ACCEPT     tcp  --  anywhere             anywhere             tcp dpt:http
REJECT     all  --  anywhere             anywhere             reject-with icmp-host-prohibited
```

60. How do you allow incoming SSH connections on iptables?

Answer: Use the iptables -A command:

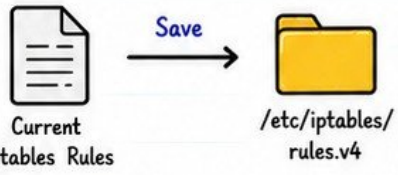
```
$ sudo iptables -A INPUT -p tcp --dport 22 -j ACCEPT
```



61. How do you save iptables rules?

Answer: Use the iptables-save command:

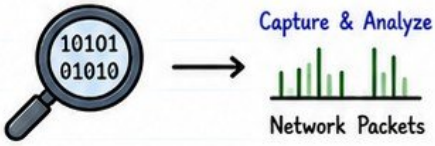
```
$ sudo iptables-save > /etc/iptables/rules.v4
```



BEST PRACTICE
Always persist iptables rules. Otherwise rules will be lost after reboot.

62. What is the tcpdump command used for?

Answer: tcpdump is a packet analyzer that captures and displays the network traffic passing through a network interface.



PRO TIP
Use tcpdump in production carefully. It can impact performance if overused.

63. How do you capture packets on a specific interface using tcpdump?

Answer: Use the tcpdump -i command:

```
$ sudo tcpdump -i eth0
```

```
$ sudo tcpdump -i eth0
tcpdump: listening on eth0, link-type EN10MB
(Ethernet), snapshot length 262144 bytes
12:01:14.123456 IP 192.168.1.10.54321 > 8.8.8.53: 51234+ A? example.com. (28)
12:01:14.223456 IP 8.8.8.53 > 192.168.1.10.54321: 51234 1/0/0 A 93.184.216.34 (44)
...
```

64. What is the ss command used for?

Answer: ss is used to dump socket statistics and display information about network connections.



- Faster than netstat
- More detailed info
- Shows TCP states
- Better for troubleshooting

65. How do you display listening ports using ss?

Answer: Use the ss -tuln command:

```
$ ss -tuln
```

```
$ ss -tuln
Netid  State   Recv-Q  Send-Q  Local Address:Port    Peer Address:Port
udp    UNCONN  0       0       0.0.0.0:53            0.0.0.0:*
tcp    LISTEN  0       128     0.0.0.0:22            0.0.0.0:*
tcp    LISTEN  0       128     0.0.0.0:80            0.0.0.0:*
tcp    LISTEN  0       128     [::]:22               [::]:*
```

66. What is the hostname command used for?

Answer: hostname displays or sets the system's hostname.

```
$ hostname
prod-web-01
```

Display

```
$ sudo hostname new-hostname
```

Set new hostname

NOTE
Changing hostname may require editing /etc/hostname too.

SENIOR ENGINEER INSIGHT: Networking issues cause more production outages than any other layer. Master these commands and you will debug faster, prevent issues earlier, and design more reliable systems.

VERIQTA
200 LINUX INTERVIEW Q&A

67. How do you change the system's hostname?

Answer: Use the hostnamectl command:

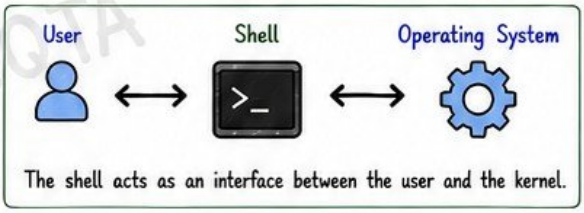
```
$ sudo hostnamectl set-hostname newhostname
```



TIP: Changing hostname may require updating /etc/hosts and DNS records in production.

68. What is a shell in Linux?

Answer: A shell is a command-line interpreter that provides a user interface for accessing the operating system's services.



PRO TIP Choose the right shell for scripting, automation, and environment consistency.

69. What is the difference between bash and sh?

Answer: bash (Bourne Again Shell) is an enhanced version of sh (Bourne Shell) with additional features like command history, tab completion, and improved scripting capabilities.

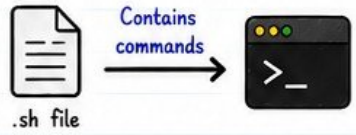
Feature	sh (Bourne Shell)	bash (Bourne Again Shell)
Basic Scripting	✓	✓
Command History	✗	✓
Tab Completion	✗	✓
Arrays	✗	✓
Advanced Features	✗	✓
Default on Most Systems	✓	✓

INSIGHT: For modern Linux distributions, bash is the default interactive shell for most users.

70. How do you create a shell script?

Answer: Create a text file with the .sh extension and add shell commands to it. For example:

```
#!/bin/bash
echo "Hello, World!"
```

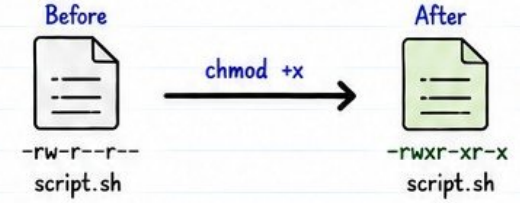


BEST PRACTICE Start every script with a shebang and set strict mode for safety.

71. How do you make a shell script executable?

Answer: Use the chmod command:

```
$ chmod +x script.sh
```



TIP: Always restrict script permissions to only what is necessary.

72. How do you run a shell script?

Answer: Use the ./ syntax:

```
$ ./script.sh
```



PRO TIP Use full path or ./ to avoid command not found errors.

73. What is the purpose of the shebang (#!) in a script?

Answer: The shebang specifies the interpreter to be used to execute the script. For example, #!/bin/bash tells the system to use Bash to run the script.

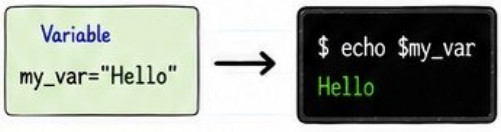


INSIGHT: Without a correct shebang, your script may run in the wrong shell.

74. How do you define a variable in a shell script?

Answer: Use the = operator without spaces:

```
variable_name="value"
```



PRO TIP Always quote variables to prevent word splitting and globbing.

SENIOR ENGINEER INSIGHT: Shell scripting is the glue of system administration and DevOps automation. Master it deeply — write safe, idempotent, and maintainable scripts for production environments.

200 LINUX INTERVIEW Q&A

75. How do you access a variable in a shell script?

Answer: Use the $ symbol:

```
$ echo $variable_name
```

Example

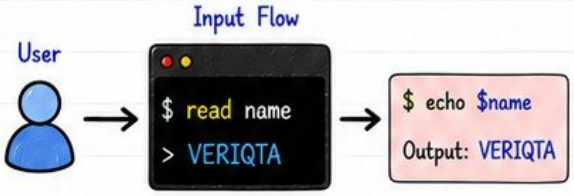
```
name="VERIQTA"
echo $name  # Output: VERIQTA
```

TIP: Always use quotes when defining variables with spaces.

76. How do you read user input in a shell script?

Answer: Use the read command:

```
$ read variable_name
```

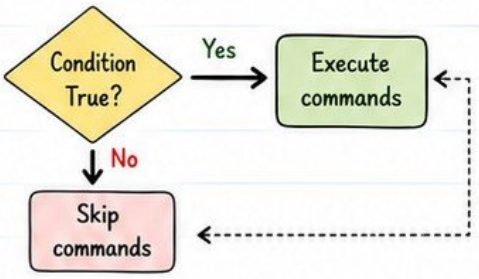


BEST PRACTICE: Always validate user input before using it in scripts.

77. How do you use an if statement in a shell script?

Answer: Use the if keyword followed by the condition and then keyword. For example:

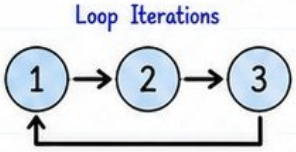
```
if [ condition ]; then
    commands
fi
```



78. How do you use a for loop in a shell script?

Answer: Use the for keyword followed by the loop variable and in keyword. For example:

```
for i in 1 2 3; do
    echo $i
done
```

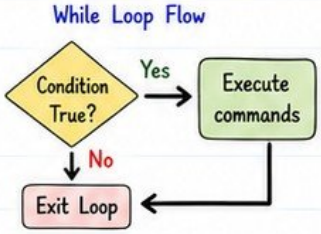


REMEMBER: Use loops for automation. Avoid repetitive manual work.

79. How do you use a while loop in a shell script?

Answer: Use the while keyword followed by the condition. For example:

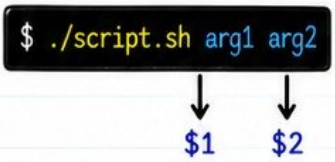
```
while [ condition ]; do
    commands
done
```



TIP: Ensure the condition changes inside the loop to avoid infinite loops.

80. How do you pass arguments to a shell script?

Answer: Use positional parameters $1, $2, etc. For example:



Inside script:

```
echo "First Arg: $1"
echo "Second Arg: $2"
```

BEST PRACTICE: Always check if required arguments are provided:

```
[ $# -lt 2 ] && exit 1
```

81. How do you display the current date and time?

Answer: Use the date command:

```
$ date
```

```
# Typical Output
Wed Jul 24 15:30:45 UTC 2024
```

USE CASES:
- Logging
- Backups
- Reports
- Monitoring

PRODUCTION NOTE: Shell scripting is the glue of system automation. Start small, test often, handle errors, and write scripts that you will be proud to run in production.

200 LINUX INTERVIEW Q&A

82. How do you display a calendar for the current month?

Answer: Use the cal command:

```
$ cal
```

Sample Output

```
      May 2024
Su Mo Tu We Th Fr Sa
            1  2  3  4
 5  6  7  8  9 10 11
12 13 14 15 16 17 18
19 20 21 22 23 24 25
26 27 28 29 30 31
```

TIP: cal can also show other months/years:
- cal 5 2025
- cal 2025

83. How do you display the current user's username?

Answer: Use the whoami command:

```
$ whoami
```

Sample Output

```
$ whoami
veriqta
```

REMEMBER: whoami is more reliable in scripts than echo $USER or echo $LOGNAME.

84. How do you display a list of users currently logged in?

Answer: Use the who command:

```
$ who
```

Sample Output

```
$ who
veriqta    pts/0    2024-05-24 08:15 (192.168.1.10)
devops     pts/1    2024-05-24 08:20 (192.168.1.25)
root       pts/2    2024-05-24 08:30 (192.168.1.30)
```

BEST PRACTICE: Use 'who' to monitor active sessions during incidents or audits.

85. How do you display information about the system?

Answer: Use the uname command. For detailed information:

```
$ uname -a
```

Sample Output

```
$ uname -a
Linux veriqta-server 5.15.0-88-generic #98-Ubuntu
SMP Wed Apr 10 12:00:00 UTC 2024 x86_64 x86_64
x86_64 GNU/Linux
```

TIP: uname helps identify kernel version, OS, architecture—critical for troubleshooting and compatibility.

86. What is the uptime command used for?

Answer: uptime displays how long the system has been running, the number of users, and the load average.

```
$ uptime
```

Sample Output

```
$ uptime
08:45:30 up 5 days,  3:12,  3 users,  load average: 0.15, 0.20, 0.25
```

How long system has been running | Number of users logged in | Load average (1m, 5m, 15m)

NOTE: Load average > CPU cores = system may be under heavy load.

87. How do you display the system's load average?

Answer: Use the uptime or top command.

Using uptime

```
$ uptime
```

Load average: 0.15, 0.20, 0.25

Using top

```
$ top
```

Load Avg: 0.15, 0.20, 0.25

REMEMBER: Consistently high load can indicate CPU, I/O, memory, or application issues.

88. What is the dmesg command used for?

Answer: dmesg displays the kernel ring buffer messages, useful for debugging hardware and driver issues.

```
$ dmesg | tail -n 10
```

Sample Output (last 3 lines)

```
[  1234.567890] eth0: Link is Up - 1000Mbps Full Duplex
[  1234.568001] IPv6: ADDRCONF(NETDEV_CHANGE): eth0: link becomes ready
[  1234.570123] EXT4-fs (sda1): mounted filesystem with ordered data mode
```

BEST PRACTICE: Always check dmesg after hardware changes or boot problems.

89. How do you clear the terminal screen?

Answer: Use the clear command:

```
$ clear
```



TIP: Ctrl + L also clears the terminal screen.

SENIOR ENGINEER INSIGHT: Master these core commands. They are the foundation of every debugging session, automation script, and production support task you will handle as an engineer.

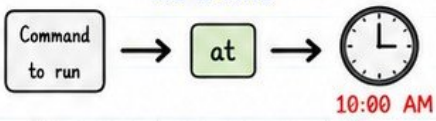
200 LINUX INTERVIEW Q&A

90. How do you schedule a one-time job using at?

Answer: Use the at command followed by the time. For example:

```
$ echo "command" | at 10:00
```

How it works



The command will run once at the specified time.

TIP:
Use 24-hour format.
Example:
13:30, 23:00
Not: 1:30 PM

91. How do you list scheduled at jobs?

Answer: Use the atq command:

```
$ atq
```

Sample Output

```
$ atq
1    Tue May 28 10:00:00 2024 a user
2    Wed May 29 14:30:00 2024 a user
```

NOTE:
The number on the left is the job ID used to remove the job.

92. How do you remove a scheduled at job?

Answer: Use the atrm command followed by the job number:

```
$ atrm job_number
```

Example

```
$ atq
1    Tue May 28 10:00:00 2024 a user
2    Wed May 29 14:30:00 2024 a user
$ atrm 1
Job 1 has been removed.
```

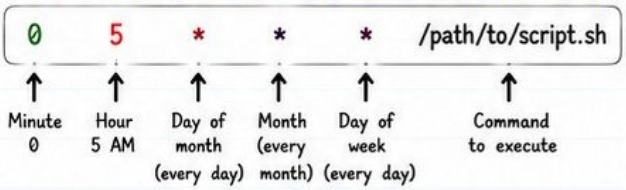
BEST PRACTICE:
Always remove unused jobs to avoid unexpected execution and confusion.

93. How do you schedule a recurring job using cron?

Answer: Edit the crontab file using crontab -e and add the job with the schedule. For example:

```
0 5 * * * /path/to/script.sh
```

Cron format (Minute Hour Day Month DayOfWeek Command)



TIP:
This runs the script every day at 5:00 AM.
Use crontab.guru to generate complex schedules.

94. How do you list all cron jobs for the current user?

Answer: Use the crontab -l command:

```
$ crontab -l
```

Sample Output

```
$ crontab -l
# m h  dom mon dow  command
0  5  *  *  *  /path/to/script.sh
*/10 * * * *  /path/to/health-check.sh
```

REMEMBER:
Cron will not run if your system is powered off.
Use a reliable server.

95. How do you remove all cron jobs for the current user?

Answer: Use the crontab -r command:

```
$ crontab -r
```

Example

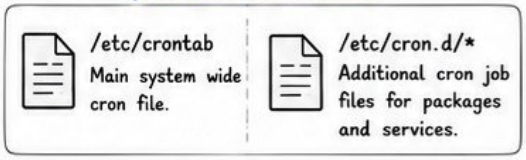
```
$ crontab -r
crontab: installing new crontab
```

BEST PRACTICE:
Double-check before running crontab -r.
It removes ALL jobs without confirmation.

96. How do you view the system's scheduled cron jobs?

Answer: View the /etc/crontab file and files in /etc/cron.d/.

System-wide cron locations



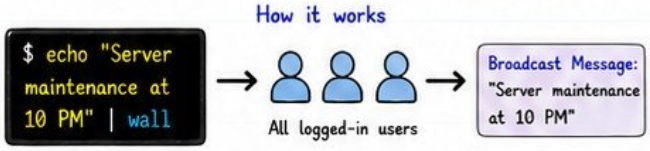
TIP:
Do not edit system files unless necessary.
Use crontab -e for user-specific jobs.

97. How do you send a message to all logged-in users?

Answer: Use the wall command:

```
$ echo "message" | wall
```

How it works



PRO TIP:
Use wall carefully.
Important alerts only.
Avoid spamming users.

SENIOR ENGINEER INSIGHT: Automation is the backbone of reliability. Use 'at' for one-time tasks, cron for recurring jobs, and always test scripts before scheduling in production.

200 LINUX INTERVIEW Q&A

98. How do you display the amount of free and used memory in the system?

Answer: Use the free command:

```
$ free -h
```

Sample Output

```
$ free -h
              total        used        free      shared  buff/cache   available
Mem:          7.7Gi       2.1Gi       1.8Gi       230Mi       3.8Gi       4.9Gi
Swap:         2.0Gi          0B       2.0Gi
```

TIP: -h shows human readable format. Watch "available" column for apps.

99. How do you check for available software updates?

Answer:

For Debian-based systems, use:

```
$ sudo apt update
```

Debian / Ubuntu



For Red Hat-based systems, use:

```
$ sudo yum check-update
```

RHEL / CentOS / Rocky / AlmaLinux



NOTE: This does NOT install updates. It only checks. what is available.

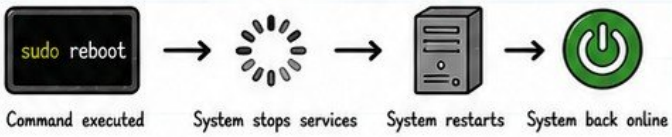
BEST PRACTICE: Always review updates, read changelogs, then apply during maintenance windows.

100. How do you reboot the system?

Answer: Use the reboot command:

```
$ sudo reboot
```

What happens when you reboot



CAUTION: Rebooting will interrupt running processes and active sessions.

101. How do you shut down the system?

Answer: Use the shutdown command:

```
$ sudo shutdown now
```

What happens when you shutdown



BEST PRACTICE: Always shutdown cleanly to avoid filesystem corruption and data loss.

102. How do you schedule a system shutdown?

Answer: Use the shutdown command with a time argument:

```
$ sudo shutdown +10
```

Example

```
$ sudo shutdown +10
Broadcast message from root@server (pts/0) (Tue May 28 10:15:00 2024):

The system will shut down for system halt in 10 minutes!
```

Shutdown in 10 minutes.

TIP: Use +m for minutes or HH:MM for specific time.

103. How do you cancel a scheduled shutdown?

Answer: Use the shutdown -c command:

```
$ sudo shutdown -c
```

Example

```
$ sudo shutdown -c
Shutdown cancelled.
```

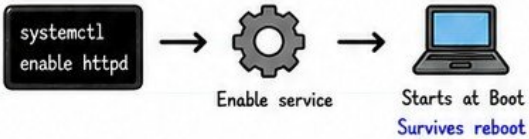
NOTE: This cancels only the last scheduled shutdown.

104. How do you enable a service to start at boot?

Answer: Use the systemctl enable command:

```
$ sudo systemctl enable service_name
```

How it works



BEST PRACTICE: Ensure critical services are enabled so they start automatically after reboot.

SENIOR ENGINEER INSIGHT: Automation + Scheduling + Monitoring = Reliable Systems.
Use 'at' for one-time tasks, 'cron' for recurring jobs, and always automate system operations safely.

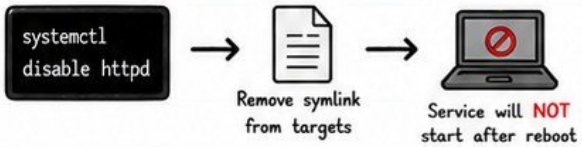
200 LINUX INTERVIEW Q&A

105. How do you disable a service from starting at boot?

Answer: Use the systemctl disable command:

```
$ sudo systemctl disable service_name
```

How it works



TIP: Use disable when you never want the service to start on boot.

106. How do you start a service?

Answer: Use the systemctl start command:

```
$ sudo systemctl start service_name
```

How it works



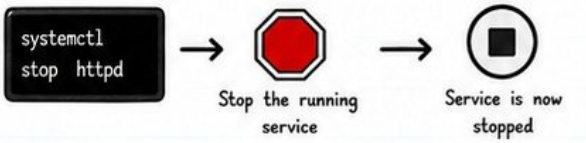
BEST PRACTICE: Start services before testing applications or performing health checks.

107. How do you stop a service?

Answer: Use the systemctl stop command:

```
$ sudo systemctl stop service_name
```

How it works



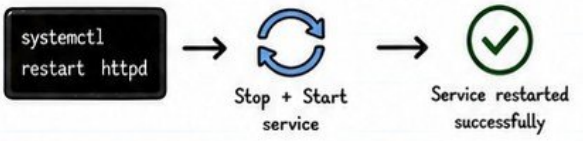
NOTE: Stopping a service does not prevent it from starting again at boot.

108. How do you restart a service?

Answer: Use the systemctl restart command:

```
$ sudo systemctl restart service_name
```

How it works



BEST PRACTICE: Use restart after configuration changes so the service picks up new settings.

109. How do you check the status of a service?

Answer: Use the systemctl status command:

```
$ systemctl status service_name
```



Shows if service is running, failed, or inactive.

Example Output: systemctl status httpd

```
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; vendor preset: enabled)
   Active: active (running) since Tue 2024-05-28 10:20:11 UTC; 2h 15min ago
Main PID: 1234 (httpd)
   Tasks: 7 (limit: 2311)
  Memory: 14.2M
  CGroup: /system.slice/httpd.service
          ├─1234 /usr/sbin/httpd -DFOREGROUND
          └─1235 /usr/sbin/httpd -DFOREGROUND

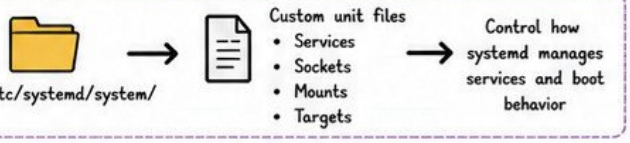
May 28 10:20:11 server systemd[1]: Starting The Apache HTTP Server...
May 28 10:20:11 server systemd[1]: Started The Apache HTTP Server.
```

TIP: Look for "Active: active (running)" for a healthy service.

110. What is the purpose of the /etc/systemd/system/ directory?

Answer: It contains systemd unit files for services and targets that are enabled to start at boot.

Directory Purpose



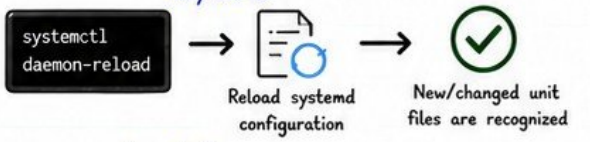
NOTE: Use this directory for custom units, overrides, and service management.

111. How do you reload systemd manager configuration?

Answer: Use the systemctl daemon-reload command:

```
$ sudo systemctl daemon-reload
```

Why use it?



BEST PRACTICE: Run this after adding or modifying unit files before starting the service.

112. What is the journalctl command used for?

Answer: journalctl is used to query and display logs from the systemd journal.

Example Usage

```
$ journalctl
$ journalctl -u httpd
$ journalctl -xe
$ journalctl --since "1 hour ago"
```

View, filter, and analyze system logs efficiently.

TIP: journalctl replaces traditional log files like /var/log/messages in modern systems.

SENIOR ENGINEER INSIGHT: Mastering systemd is essential for real-world Linux administration. Know your services, automate smartly, and always build reliable, observable systems.

200 LINUX INTERVIEW Q&A

113. How do you display the latest system logs using journalctl?

Answer: Use the journalctl -e command:

```
$ journalctl -e
```

What it does

Shows the most recent logs and updates in real time (live stream).



TIP: Press Ctrl + C to exit the live log stream.

114. How do you display kernel logs using journalctl?

Answer: Use the journalctl -k command:

```
$ journalctl -k
```

What it does

Displays logs from the kernel (useful for boot, driver, and hardware issues).

```
$ journalctl -k -n 5
May 28 12:10:01 kernel: Linux version 5.15.0-97-generic
May 28 12:10:01 kernel: Command line: BOOT_IMAGE=/vmlinuz...
May 28 12:10:01 kernel: CPU: Intel(R) Xeon(R) CPU..
May 28 12:10:01 kernel: ACPI: Core revision 20220331
May 28 12:10:02 kernel: PCI host bridge to bus 0000:00
```

NOTE: Kernel logs help debug boot failures, panic, driver errors, and hardware issues.

115. How do you display logs for a specific service using journalctl?

Answer: Use the journalctl -u command:

```
$ journalctl -u service_name
```

Example

View logs for nginx service:

```
$ journalctl -u nginx -n 10 --no-pager
May 28 12:20:01 server nginx[1234]: Starting NGINX...
May 28 12:20:01 server nginx[1234]: Listening on 0.0.0.0:80
May 28 12:20:01 server nginx[1234]: Worker process started
May 28 12:20:05 server nginx[1234]: 200 OK GET /index.html
```

BEST PRACTICE: Use -u to filter logs by service. Combine with -f for live follow.

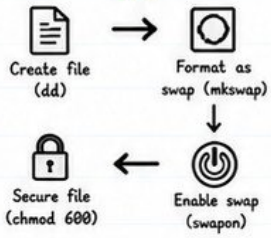
116. How do you set up a swap file in Linux?

Answer: Use the dd, mkswap, and swapon commands:

Steps

1. `$ sudo dd if=/dev/zero of=/swapfile bs=1M count=2048`
2. `$ sudo mkswap /swapfile`
3. `$ sudo swapon /swapfile`
4. `$ sudo chmod 600 /swapfile`

Flow



TIP: Swap should be on fast storage. SSD is preferred.

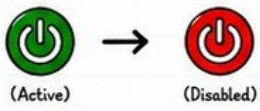
117. How do you disable a swap file?

Answer: Use the swapoff command:

```
$ sudo swapoff /swapfile
```

What it does

Turns off the swap file immediately.



NOTE: Swapoff frees the swap space and stops using it.

118. How do you permanently enable a swap file?

Answer: Add the swap file entry to /etc/fstab:

Edit /etc/fstab

```
/swapfile   swap   swap   defaults   0   0
```

Swap file path | Type (swap) | Filesystem type | Options | Dump (0) | Pass (0)

BEST PRACTICE: Verify with swapon -a after reboot to ensure it mounts automatically.

119. How do you check swap usage?

Answer: Use the swapon -s or free -h command:

```
$ swapon -s
```

or

```
$ free -h
```

Example Output: swapon -s

Filename	Type	Size	Used	Priority
/swapfile	file	2G	0B	-2

Example Output: free -h

	total	used	free	shared	buff/cache	available
Mem:	7.7Gi	2.1Gi	3.0Gi	230Mi	2.6Gi	5.1Gi
Swap:	2.0Gi	128Mi	1.9Gi			

TIP: If used swap keeps growing, check for memory leaks or increase RAM.

PRODUCTION NOTE: Systemd + journald + swap are esential for a stable Linux environment. Automate monitoring, review logs proactively, and manage resources efficiently. AUTOMATE EVERYTHING

200 LINUX INTERVIEW Q&A

120. What is the df command used for?

Answer: df displays the amount of disk space used and available on file systems.

Example: df -h

```
$ df -h
Filesystem      Size  Used Avail Use% Mounted on
/dev/sda1        50G   18G   30G  38% /
/dev/sda2       100G   40G   60G  48% /home
tmpfs           2.0G     0  2.0G   0% /run
/dev/sda3       200G  120G   80G  60% /data
```

TIP: Use -h for human readable sizes. Use -T to see filesystem types.

121. What is the du command used for?

Answer: du estimates file space usage and displays the disk usage of files and directories.

Example: du -sh *

```
$ du -sh *
4.0K    file1.txt
12M     documents/
450M    videos/
2.1G    backup/
```

NOTE: du scans files and directories to calculate usage, so it may be slow on large trees.

122. How do you check the disk usage of a specific directory?

Answer: Use the du -sh command:

```
$ du -sh /path/to/directory
```

Example Output

```
$ du -sh /var/log
320M    /var/log
```

BEST PRACTICE: Use -h for human readable and -c to show total.

123. How do you create a symbolic link?

Answer: Use the ln -s command:

```
$ ln -s /path/to/target /path/to/link
```

Symbolic Link (Soft Link)



TIP: Symlinks are like shortcuts. If target is deleted, symlink becomes broken.

124. How do you create a hard link?

Answer: Use the ln command:

```
$ ln /path/to/target /path/to/link
```

Hard Link



NOTE: Hard links are another name for the same data on disk (same inode).

125. What is the difference between a symbolic link and a hard link?

Answer: A symbolic link points to a file or directory by name, whereas a hard link points directly to the inode of a file. Symbolic links can cross file system boundaries, while hard links cannot.

Comparison

Feature	Symbolic Link (Soft Link)	Hard Link
Points to	File/Directory path	Inode of the file
Cross Filesystem	Yes	No
Broken if target deleted?	Yes (broken link)	No (data still exists)
Can link directories?	Yes	No (usually not)
Use case	Shortcuts, convenience	Backup, redundancy

BEST PRACTICE: Use symlinks for directories and cross-filesystem links. Use hard links for files within the same filesystem.

126. How do you view the contents of a directory?

Answer: Use the ls command:

```
$ ls /path/to/directory
```

Example Output

```
$ ls /etc
bash.bashrc     host.conf     passwd     fstab
ssl/            nginx/        shadow     group
```

TIP: Add -l for long list and -h for human readable sizes.

127. How do you view hidden files in a directory?

Answer: Use the ls -a command:

```
$ ls -a /path/to/directory
```

Example Output

```
$ ls -a /home/user
.   ..   .bashrc   .config   file.txt   .ssh   .profile
```

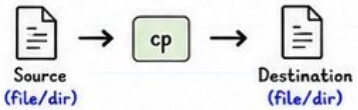
NOTE: Files starting with . (dot) are hidden.

128. How do you copy files?

Answer: Use the cp command.

```
$ cp file.txt /backup/          # copy file
$ cp -r /dir1 /backup/          # copy directory recursively
$ cp -v file1 file2             # verbose copy
$ cp -p file1 file2             # preserve mode, ownership, timestamps
```

Copy Flow



BEST PRACTICE: Use -r for directories and -p to preserve permissions and timestamps.

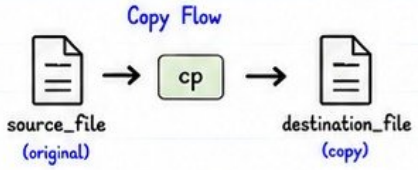
SENIOR ENGINEER INSIGHT: Understanding disk usage, links, and system logs is crucial for troubleshooting, performance tuning, and maintaining a stable Linux environment in production.

200 LINUX INTERVIEW Q&A

128. How do you copy files?

Answer: Use the cp command:

```
$ cp source_file destination_file
```

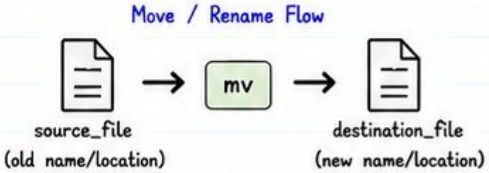


TIP: Use -r for directories and -p to preserve permissions.

129. How do you move or rename files?

Answer: Use the mv command:

```
$ mv source_file destination_file
```

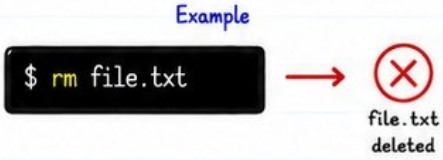


NOTE: mv can move files across directories or rename within the same directory.

130. How do you delete files?

Answer: Use the rm command:

```
$ rm filename
```

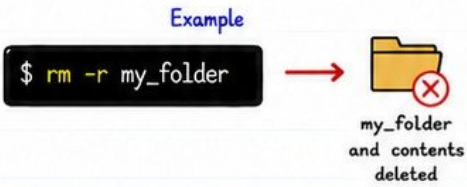


TIP: No confirmation by default. Be careful!

131. How do you delete directories and their contents?

Answer: Use the rm -r command:

```
$ rm -r directory_name
```

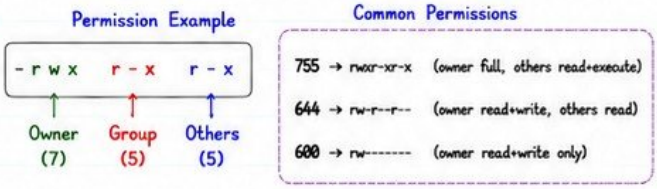


NOTE: Use rm -rf only when you are 100% sure. It cannot be undone.

132. How do you change file permissions?

Answer: Use the chmod command:

```
$ chmod 755 filename
```



BEST PRACTICE: Follow least privilege principle. Don't give excess permissions.

133. How do you change file ownership?

Answer: Use the chown command:

```
$ sudo chown user:group filename
```

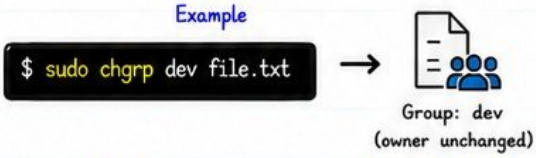


TIP: Use chown recursively with -R for directories. Example: chown -R alice:dev /data

134. How do you change the group ownership of a file?

Answer: Use the chgrp command:

```
$ sudo chgrp group filename
```



NOTE: chgrp does not change the user (owner), only the group.

135. How do you find files by name?

Answer: Use the find command:

```
$ find /path/to/search -name "filename"
```

Example

```
$ find /home/user -name "file.txt"
/home/user/file.txt
/home/user/docs/file.txt
/home/user/backup/file.txt
```

Useful find Options

```
-name      → match by name
-type f    → search files
-type d    → search directories
-mtime -7  → modified in last 7 days
```

BEST PRACTICE: Use quotes around filename to avoid shell expansion.

136. How do you search for files containing a specific string?

Answer: Use the grep -r command:

```
$ grep -r "search_string" /path/to/search
```

Example

```
$ grep -r "ERROR" /var/log
/var/log/syslog:May 28 12:10:01 ERROR Disk full
/var/log/app.log:May 28 12:11:22 ERROR Failed to connect
/var/log/nginx/error.log:2024/05/28 ERROR 502 Bad Gateway
```

TIP: Use -i for case insensitive search, --color=auto for highlighted results.

SENIOR ENGINEER INSIGHT: Mastering file management, permissions, and search tools is essential for system administration, troubleshooting, and automation.

200 LINUX INTERVIEW Q&A

137. What is the locate command used for?

Answer: locate searches for files in a pre-built database, making it faster than find.

Example

```
$ locate nginx.conf
/etc/nginx/nginx.conf
/usr/share/doc/nginx/examples/nginx.conf
/var/lib/docker/overlay2/.../nginx.conf
```

TIP: locate is very fast but may return outdated results if the database is not updated.

138. How do you update the locate database?

Answer: Use the updatedb command:

```
$ sudo updatedb
```

What it does

Scans the filesystem and updates the locate database (usually runs via cron daily).



NOTE: updatedb can take time on large systems. Run during low usage hours.

139. How do you view the manual page for a command?

Answer: Use the man command:

```
$ man command_name
```

Example

```
$ man ls
LS(1)              User Commands              LS(1)

NAME
       ls - list directory contents
SYNOPSIS
       ls [OPTION]... [FILE]...
DESCRIPTION
       List information about the FILEs (the current
       directory by default).
```

TIP: Press q to quit the manual page. Navigate with arrow keys, / to search.

140. How do you view the help information for a command?

Answer: Use the --help option:

```
$ command_name --help
```

Example

```
$ ls --help
Usage: ls [OPTION]... [FILE]...
List information about the FILEs (the
current directory by default).
...
```

NOTE: --help gives a quick summary of options and usage.

141. How do you display the current environment variables?

Answer: Use the env command:

```
$ env
```

Example Output

```
USER=veriqta
HOME=/home/veriqta
SHELL=/bin/bash
PATH=/usr/local/bin:/usr/bin:/bin
LANG=en_US.UTF-8
...
```

BEST PRACTICE: Environment variables control system behavior and application configuration.

142. How do you set an environment variable?

Answer: Use the export command:

```
$ export VAR_NAME=value
```

Example

```
$ export APP_ENV=production
$ echo $APP_ENV
production
```

Current Shell

export APP_ENV=production → Used by commands and scripts

TIP: Variables set this way exist only in the current shell session.

143. How do you add a directory to the PATH variable?

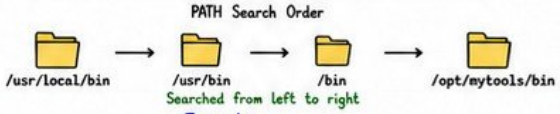
Answer: Use the export command to modify PATH:

```
$ export PATH=$PATH:/path/to/directory
```

Example

```
$ export PATH=$PATH:/opt/mytools/bin
$ echo $PATH
/usr/local/bin:/usr/bin:/bin:/opt/mytools/bin
```

PATH Search Order



BEST PRACTICE: Add directories to PATH to run your custom commands from anywhere.

144. What is the alias command used for?

Answer: alias creates shortcuts for commands. For example:

```
$ alias ll='ls -l'
```

Example

```
$ alias ll='ls -l'
$ ll /etc
total 36
drwxr-xr-x   2 root root  4096 May 28 10:30 nginx
-rw-r--r--   1 root root   220 May 28 10:30 hosts
-rw-r--r--   1 root root   103 May 28 10:30 resolv.conf
```

NOTE: Add aliases to ~/.bashrc or ~/.bash_aliases to make them persistent.

SENIOR ENGINEER INSIGHT: Mastering these commands improves productivity, automates tasks, and helps you troubleshoot systems like a real Linux engineer.

200 LINUX INTERVIEW Q&A

145. How do you check CPU information in Linux?

Answer: Use the lscpu command:

```
$ lscpu
```

Example Output

```
Architecture:          x86_64
CPU op-mode(s):        32-bit, 64-bit
Byte Order:            Little Endian
CPU(s):                8
On-line CPU(s) list:   0-7
Thread(s) per core:    2
Core(s) per socket:    4
Socket(s):             1
Model name:            Intel(R) Core(TM) i7-8650U CPU @ 1.90GHz
CPU MHz:               2112.000
L1d cache:             128 KiB (4 instances)
L1i cache:             128 KiB (4 instances)
L2 cache:              1 MiB (4 instances)
L3 cache:              8 MiB (1 instance)
```

TIP: lscpu is a quick way to get CPU architecture, cores, threads, and cache information.

145. How do you view detailed memory information?

Answer: Use the /proc/meminfo file:

```
$ cat /proc/meminfo
```

Example Output

```
MemTotal:       16319344 kB
MemFree:         5301240 kB
MemAvailable:   10123456 kB
Buffers:          254680 kB
Cached:          6212340 kB
SwapCached:            0 kB
Active:          4321500 kB
Inactive:        2754300 kB
SwapTotal:       2097148 kB
SwapFree:        2097148 kB
```

NOTE: /proc/meminfo provides real-time memory usage details including cache, buffers, and swap.

147. How do you display block devices and partitions?

Answer: Use the lsblk command:

```
$ lsblk
```

Example Output

```
NAME   MAJ:MIN RM   SIZE RO TYPE MOUNTPOINT
sda      8:0    0   238G  0 disk
├─sda1   8:1    0   512M  0 part /boot/efi
├─sda2   8:2    0     1G  0 part /boot
└─sda3   8:3    0 236.5G  0 part /
sdb      8:16   0   1.8T  0 disk
└─sdb1   8:17   0   1.8T  0 part /mnt/data
```

BEST PRACTICE: lsblk shows block devices in a clear tree format. Use -f to show filesystem information.

148. How do you list all PCI devices?

Answer: Use the lspci command:

```
$ lspci
```

Example Output

```
00:00.0 Host bridge: Intel Corporation 8th Gen Core Processor
00:02.0 VGA compatible controller: Intel Corporation UHD Graphics 620
00:14.0 USB controller: Intel Corporation Cannon Lake PCH USB 3.1 xHCI
00:1f.3 Audio device: Intel Corporation Cannon Lake PCH cAVS
01:00.0 Ethernet controller: Realtek Semiconductor Co., Ltd. RTL8111/8168
```

TIP: lspci is useful for troubleshooting hardware issues and verifying driver support.

149. How do you list all USB devices?

Answer: Use the lsusb command:

```
$ lsusb
```

Example Output

```
Bus 002 Device 002: ID 8087:0024  Intel Corp. Integrated Rate Matching Hub
Bus 002 Device 003: ID 046d:c52b  Logitech, Inc. Unifying Receiver
Bus 002 Device 004: ID 0781:5580  SanDisk Corp. Ultra Fit
Bus 002 Device 001: ID 1d6b:0002  Linux Foundation 2.0 root hub
```

NOTE: lsusb lists all USB devices connected to your system.

150. How do you display information about attached storage devices?

Answer: Use the blkid command:

```
$ sudo blkid
```

Example Output

```
/dev/sda1: UUID="A1B2-C3D4" TYPE="vfat" PARTUUID="1234-01"
/dev/sda2: UUID="b5e6a7f8-1234-5678-90ab-cdef12345678" TYPE="ext4"
/dev/sda3: UUID="9a8b7c6d-1234-5678-90ab-fedcba987654" TYPE="ext4"
/dev/sdb1: UUID="12345678-9abc-def0-1234-56789abcdef0" TYPE="ext4"
```

BEST PRACTICE: blkid shows UUIDs and filesystem types. Useful for mounting in /etc/fstab using UUIDs.

151. How do you display hardware information for a Linux system?

Answer: Use the lshw command:

```
$ sudo lshw
```

Example Output

```
*-system
     description: Laptop
     product: ThinkPad T480s
     vendor: LENOVO
     version: ThinkPad T480s
*-cpu
     product: Intel(R) Core(TM) i7-8650U CPU @ 1.90GHz
     ├─ size: 1900MHz
     └─ capacity: 4200MHz
*-memory
     description: System Memory
        size: 15GiB
```

TIP: lshw provides a complete overview of hardware including CPU, memory, disks, network, and more.

PRODUCTION NOTE: Hardware visibility is essential for monitoring, capacity planning, and troubleshooting. These commands help you understand your system deeply.

KNOW YOUR SYSTEM. TROUBLESHOOT LIKE A PRO.

200 LINUX INTERVIEW Q&A

152. How do you display CPU usage in real time?

Answer: Use the top command:

```
$ top
```

Example Output (top)

```
top - 11:25:35 up 1 day,  2:15,  2 users,  load average: 0.15, 0.23, 0.20
Tasks: 192 total,   2 running, 190 sleeping,   0 stopped,   0 zombie
%Cpu(s):  1.2 us,  0.7 sy,  0.0 ni, 97.6 id,  0.2 wa,  0.1 hi,  0.2 si,  0.0 st
KiB Mem : 16319344 total,  5321240 free,  6212340 used,  4803764 buff/cache
KiB Swap:  2097148 total,  2097148 free,        0 used.  9729472 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR S  %CPU %MEM     TIME+ COMMAND
 2345 veriqta   20   0 1623440 211540  89320 S   2.3  1.3   0:10.25 chrome
 1234 root      20   0  434560  54812  23248 S   1.0  0.3   0:02.41 systemd
 2456 veriqta   20   0  987652  93400  41200 S   0.7  0.6   0:05.13 code
    1 root      20   0  170932  12032   8448 S   0.0  0.1   0:01.75 systemd
```

TIP: Press 'q' to quit top and return to the shell.

153. How do you monitor system resources with an interactive interface?

Answer: Use the htop command:

```
$ htop
```

Example Output (htop)

```
1  [||||||                    12.5%]   Tasks: 192, 432 thr, 2 running
2  [|||||                      8.1%]   Load average: 0.15 0.23 0.20
3  [||||                       6.2%]   Uptime: 1 day, 02:15:35
4  [|||||                      9.0%]
Mem[|||||||||||||||| 5.32G/15.6G]
Swp[                    0K/2.00G]

  PID USER      PRI  NI    VIRT    RES    SHR S CPU% MEM%   TIME+  Command
 2345 veriqta    20   0 1623440 211540  89320 S  2.3  1.3  0:10.25 chrome
 1234 root       20   0  434560  54812  23248 S  1.0  0.3  0:02.41 systemd
 2456 veriqta    20   0  987652  93400  41200 R  0.7  0.6  0:05.13 code
 5678 veriqta    20   0  525432  61200  25200 S  0.4  0.4  0:01.33 docker
```

NOTE: htop is more user-friendly than top. Use arrow keys to navigate.

154. How do you display virtual memory statistics?

Answer: Use the vmstat command:

```
$ vmstat
```

Example Output (vmstat 1 5)

```
procs -----------memory---------- ---swap-- -----io---- -system-- ------cpu-----
 r  b   swpd   free   buff  cache   si   so    bi    bo   in   cs us sy id wa st
 1  0      0 5321240 256468 4803764    0    0    10    22  120  230  1  1 97  0  1
 0  0      0 5318200 256468 4804120    0    0     0     0  115  220  1  0 98  0  1
 0  0      0 5318120 256468 4804200    0    0     0     0  110  210  1  0 98  0  1
 0  0      0 5319000 256432 4804000    0    0     0     8  118  225  1  1 97  0  1
 0  0      0 5318500 256468 4804500    0    0     0    12  122  235  1  1 97  0  1
```

TIP: vmstat 1 5 shows stats every 1 second for 5 intervals.

155. How do you monitor disk I/O statistics?

Answer: Use the iostat command:

```
$ iostat
```

Example Output (iostat -xz 1 2)

```
Device            r/s     w/s     rkB/s     wkB/s   avgrq-sz avgqu-sz   await  svctm  %util
sda              1.25    0.50     32.00      8.00     25.60     0.01    1.20   0.50   0.09
sdb              0.10    2.30     10.00    120.00     52.00     0.03    5.60   1.10   0.25
nvme0n1          0.50    1.00     20.00     50.00     40.00     0.02    2.30   0.70   0.15
```

BEST PRACTICE: Use -xz for extended stats. Helps identify I/O bottlenecks.

156. How do you monitor real-time disk input and output activity?

Answer: Use the iotop command:

```
$ sudo iotop
```

Example Output (iotop)

```
Total DISK READ : 12.34 M/s | Total DISK WRITE : 45.67 M/s
  PID  PRIO  USER     DISK READ  DISK WRITE  SWAPIN     IO>  COMMAND
 2345  be/4  veriqta    1.23 M/s   20.45 M/s  0.00 %   65.2  java
 1234  be/4  root       0.45 M/s   10.00 M/s  0.00 %   25.1  dockerd
 3456  be/4  veriqta    0.12 M/s    5.00 M/s  0.00 %   10.7  postgres
 5678  be/4  veriqta    0.05 M/s    1.20 M/s  0.00 %    2.4  rsync
```

NOTE: Requires root privileges. Shows processes using disk I/O.

157. How do you display process IDs and resource usage in a tree format?

Answer: Use the pstree command:

```
$ pstree
```

Example Output (pstree)

```
systemd─┬─sshd───bash───vim
        ├─cron
        ├─NetworkManager───dnsmasq
        ├─docker─┬─dockerd───containerd───containerd-shim───nginx
        │        └─(sd-pam)
```

TIP: pstree -p shows PIDs next to process names.

158. How do you display detailed information about a running process?

Answer: Use the ps command with the -p option:

```
$ ps -p PID -f
```

Example Output (ps -p 2345 -f)

```
UID          PID    PPID  C STIME TTY          TIME CMD
veriqta     2345    1234  2 10:15 ??       00:00:10 /usr/bin/java -jar
                                                    /opt/app/app.jar
```

UID	: Owner of the process	C	: CPU usage
PPID	: Parent Process ID	TTY	: Terminal
STIME	: Start time	TIME	: CPU time used

BEST PRACTICE: Use ps -ef for all processes and ps aux for a human-readable list.

PRODUCTION NOTE: Monitoring system resources helps you identify performance issues, prevent outages, and optimize infrastructure for production workloads.

MONITOR TODAY, PREVENT OUTAGES TOMORROW.

200 LINUX INTERVIEW Q&A

159. How do you monitor CPU usage for each processor core?

Answer: Use the mpstat command:

```
$ mpstat -P ALL
```

Example Output (mpstat -P ALL 1 5)

```
Linux 5.15.0-88-generic (veriqta)     05/28/2025     _x86_64_     (4 CPU)

CPU      %usr   %nice   %sys   %iowait   %irq   %soft   %steal   %idle
all     12.35    0.02   3.45     0.15   0.00    0.28    0.00   83.75
0       15.20    0.00   4.10     0.10   0.00    0.30    0.00   80.30
1       11.50    0.00   3.20     0.20   0.00    0.20    0.00   84.90
2       11.20    0.00   3.00     0.10   0.00    0.30    0.00   85.40
3       11.50    0.00   3.50     0.20   0.00    0.30    0.00   84.50
```

TIP: mpstat shows per-CPU statistics. Use interval count to refresh multiple times.

160. How do you display process activity over time?

Answer: Use the pidstat command:

```
$ pidstat
```

Example Output (pidstat 1 5)

```
Linux 5.15.0-88-generic      05/28/2025      _x86_64_      (4 CPU)
Time        UID       PID    %usr %system  %guest   %CPU   Command
11:30:01   1000      2345    1.02    0.51    0.00   1.53   java
11:30:01      0         1    0.00    0.20    0.00   0.20   systemd
11:30:01   1000      3456    0.50    0.30    0.00   0.80   chrome
11:30:01      0         2    0.00    0.00    0.00   0.00   kthreadd
...
```

NOTE: pidstat helps you monitor individual process performance over time.

161. How do you monitor network bandwidth usage in real time?

Answer: Use the iftop command:

```
$ sudo iftop
```

Example Output (iftop)

```
                          12.5Kb      25.0Kb      37.5Kb      50.0Kb      62.5Kb
192.168.1.10      => 93.184.216.34                7.21Kb      6.71Kb      6.25Kb
192.168.1.10      => 142.250.72.78                2.31Kb      2.10Kb      1.98Kb
192.168.1.10      => 151.101.1.69                 1.12Kb      1.05Kb      0.98Kb
192.168.1.10      => 172.217.16.78                0.45Kb      0.40Kb      0.38Kb
192.168.1.10      => 104.16.248.249               0.32Kb      0.28Kb      0.26Kb
TX:       cum: 15.4KB   peak: 15.4Kb   rates: 15.4Kb   15.4Kb   15.4Kb
RX:       cum:  3.2KB   peak:  3.2Kb   rates:  3.2Kb    3.2Kb    3.2Kb
TOTAL:    cum: 18.6KB   peak: 18.6Kb   rates: 18.6Kb   18.6Kb   18.6Kb
```

BEST PRACTICE: iftop shows source, destination, and bandwidth in real time. Requires root privileges.

162. How do you monitor network traffic by process?

Answer: Use the nethogs command:

```
$ sudo nethogs
```

Example Output (nethogs)

```
PID   USER      PROGRAM          DEV      SENT         RECEIVED
2345  veriqta   java             eth0     34.5 KB/s    120.5 KB/s
3456  veriqta   chrome           eth0     12.3 KB/s     85.2 KB/s
4567  veriqta   docker-proxy     eth0      5.1 KB/s     10.3 KB/s
5678  veriqta   ssh              eth0      1.2 KB/s      2.6 KB/s
Total                                    53.1 KB/s    218.6 KB/s
```

NOTE: nethogs groups network usage by process instead of host or connection.

163. How do you display open files for a process?

Answer: Use the lsof command:

```
$ lsof -p PID
```

Example Output (lsof -p 2345)

```
COMMAND  PID   USER   FD   TYPE  DEVICE SIZE/OFF   NODE NAME
java    2345  veriqta cwd   DIR     8,1     4096    131072 /home/veriqta/app
java    2345  veriqta txt   REG     8,1    20480    524288 /usr/bin/java
java    2345  veriqta mem   REG     8,1  1342176    131089 /lib/x86_64-linux-gnu/libc.so.6
java    2345  veriqta mem   REG     8,1   155648    131200 /usr/lib/jvm/lib/server/libjvm.so
java    2345  veriqta 4u    IPv4 123456      0t0       TCP 192.168.1.10:54321->93.184.
                                                          216.34:80 (ESTABLISHED)
...
```

TIP: lsof lists all open files, sockets, pipes, and devices used by a process.

164. How do you list all open network ports and the processes using them?

Answer: Use the lsof command with the -i option:

```
$ sudo lsof -i
```

Example Output (lsof -i)

```
COMMAND   PID   USER   FD   TYPE DEVICE SIZE/OFF NODE NAME
sshd     1234   root    3u  IPv4  98765      0t0  TCP *:22 (LISTEN)
java     2345  veriqta  4u  IPv4 123456      0t0  TCP 192.168.1.10:54321->93.184.216.34:80
postgres 3456  postgres 5u  IPv4 234567      0t0  TCP 127.0.0.1:5432 (LISTEN)
docker    567   root    6u  IPv4 345678      0t0  TCP *:2375 (LISTEN)
nginx     789  www-data 7u  IPv4 456789      0t0  TCP *:80 (LISTEN)
...
```

BEST PRACTICE: Use -i to see all network connections and the process using each port.

165. How do you identify which process is using a specific TCP port?

Answer: Use the fuser command:

```
$ sudo fuser 8080/tcp
```

Example Output (fuser 8080/tcp)

```
8080/tcp:             2345
                      2345  ← Process ID using port 8080

2345/tcp:
```

NOTE: fuser quickly shows the PID of the process using a specific port.

PRODUCTION INSIGHT: System monitoring commands help you detect issues early, optimize performance, and troubleshoot problems before they impact users.

MONITOR, ANALYZE, AND KEEP SYSTEMS HEALTHY!

200 LINUX INTERVIEW Q&A

166. How do you identify the executable path of a command?

Answer: Use the which command:

```
$ which command_name
```

Example Output

```
$ which ls
/usr/bin/ls

$ which python3
/usr/bin/python3
```

TIP: which returns the first executable found in your $PATH.

167. How do you display all locations of a command, source, and manual page?

Answer: Use the whereis command:

```
$ whereis command_name
```

Example Output

```
$ whereis gcc
gcc: /usr/bin/gcc /usr/share/man/man1/gcc.1.gz
     /usr/share/info/gcc.info.gz
```

NOTE: whereis searches for the binary, source files, and manual page locations.

168. How do you determine the type of a file?

Answer: Use the file command:

```
$ file filename
```

Example Output

```
$ file /etc/passwd
/etc/passwd: ASCII text

$ file /bin/ls
/bin/ls: ELF 64-bit LSB pie executable,
x86-64, version 1 (GNU/Linux), dynamically linked,
interpreter /lib64/ld-linux-x86-64.so.2
```

BEST PRACTICE: file is very useful for identifying unknown or suspicious files.

169. How do you compare two text files line by line?

Answer: Use the diff command:

```
$ diff file1 file2
```

Example Output

```
$ diff file1.txt file2.txt
3c3
< Linux is powerful
---
> Linux is awesome
5d4
< Learning never stops
```

TIP: diff shows the differences in a compact format. Great for scripts and quick checks.

170. How do you display the differences between two files side by side?

Answer: Use the sdiff command:

```
$ sdiff file1 file2
```

Example Output (sdiff file1.txt file2.txt)

```
file1.txt                    | file2.txt
1 Linux is powerful          | 1 Linux is awesome
2 Automation saves time      | 2 Automation saves time
3 DevOps improves delivery   | 3 DevOps improves delivery
4 Learning never stops       | 4 Consistency builds trust

Use ⟷ to navigate | q to quit
```

NOTE: sdiff provides a side-by-side view. Navigate with arrow keys.

171. How do you compare and merge files interactively?

Answer: Use the vimdiff command:

```
$ vimdiff file1 file2
```

Example Output (vimdiff file1.txt file2.txt)

```
file1.txt                    | file2.txt
1 Linux is powerful          | 1 Linux is awesome
2 Automation saves time      | 2 Automation saves time
3 DevOps improves delivery   | 3 DevOps improves delivery
4 Learning never stops       | 4 Consistency builds trust
~                            | ~
~                            | ~
[No Name]  4,1  All | [No Name]  4,1  All
    Use ]c / [c to jump between differences
    :diffget to take change from other file
    :diffput  to put change to other file
    :q to quit vimdiff
```

BEST PRACTICE: vimdiff is perfect for merging configuration files, code, and important documents interactively.

172. How do you calculate and verify a file's SHA-256 checksum?

Answer: Use the sha256sum command:

```
$ sha256sum filename
```

Example Output

```
$ sha256sum file.iso
98d6e1f7c6b1e0c9d5e8a3f6b2c7a1d4e5f3b0c9a8d7e6f5b4c3d2a1e0f9d8c  file.iso

<---------- SHA-256 Checksum----------> Filename
```

TIP: Use sha256sum to verify file integrity after downloads or transfers.

PRODUCTION INSIGHT: System tools and monitoring commands help you understand performance, troubleshoot issues, and maintain a healthy, secure Linux environment.

OBSERVE, ANALYZE, OPTIMIZE, AUTOMATE!

200 LINUX INTERVIEW Q&A

173. How do you calculate the MD5 checksum of a file?

Answer: Use the md5sum command:

```
$ md5sum filename
```

Example Output

```
$ md5sum file.iso
5d41402abc4b2a76b9719d911017c592   file.iso
```

TIP: MD5 is fast but not recommended for security purposes. Use SHA-256 or SHA-1 instead.

174. How do you calculate the SHA-1 checksum of a file?

Answer: Use the sha1sum command:

```
$ sha1sum filename
```

Example Output

```
$ sha1sum file.iso
b3e574c6e97fd29edaf9b1b37b5ea18d980225d3  file.iso
```

TIP: SHA-1 is more secure than MD5 but SHA-256 is preferred today.

175. How do you sort the contents of a text file?

Answer: Use the sort command:

```
$ sort filename
```

Example Output

```
$ cat names.txt
zebra
apple
banana
orange
grape
```

→

```
$ sort names.txt
apple
banana
grape
orange
zebra
```

BEST PRACTICE: sort is very useful for organizing data before further processing.

176. How do you remove duplicate lines from a sorted file?

Answer: Use the uniq command:

```
$ uniq filename
```

Example Output

```
$ cat sorted.txt
zpbra
apple
banana
banana
grape
orange
orange
```

→

```
$ uniq sorted.txt
apple
banana
grape
orange
```

NOTE: uniq removes adjacent duplicate lines. Make sure the file is sorted first.

177. How do you count the occurrence of duplicate lines in a file?

Answer: Use the uniq -c command:

```
$ sort filename | uniq -c
```

Example Output

```
$ cat names.txt
apple
apple
banana
banana
grape
grape
orange
```

→

```
$ sort names.txt | uniq -c
   2    apple
   3    banana
   2    grape
   1    orange
```

TIP: This helps in quickly finding duplicates and their frequency.

178. How do you extract specific columns from a text file?

Answer: Use the cut command:

```
$ cut -d':' -f1 /etc/passwd
```

Example Output

```
$ head -n 5 /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
```

→

```
$ cut -d':' -f1 /etc/passwd|head -5
root
daemon
bin
sys
sync
```

BEST PRACTICE: cut is powerful for parsing delimited files like CSV, /etc/passwd, logs, etc.

179. How do you join two files based on a common field?

Answer: Use the join command:

```
$ join file1 file2
```

Example Output

```
$ cat users.txt
1001  alice
1002  bob
1003  charlie
1004  david
```

```
$ cat details.txt
1002  linux
1003  devops
1005  cloud
```

→

```
$ join users.txt details.txt
1002  bob      linux
1003  charlie  devops
```

NOTE: Both files must be sorted on the join field (default is the first column).

PRODUCTION INSIGHT: File analysis and data processing commands help you audit systems, verify data integrity, parse logs, and automate real-world tasks efficiently.

AUTOMATE. ANALYZE. VERIFY. DELIVER.

200 LINUX INTERVIEW Q&A

180. How do you split a large file into smaller files?

Answer: Use the split command:

```
$ split -b 100M largefile.txt part_
```

Example Output

```
$ split -b 100M largefile.txt part_
$ ls -lh part_*
-rw-r--r--  1 user user  100M May 28 10:12 part_aa
-rw-r--r--  1 user user  100M May 28 10:12 part_ab
-rw-r--r--  1 user user  100M May 28 10:12 part_ac
-rw-r--r--  1 user user   27M May 28 10:12 part_ad
4 files, 327M total
```

TIP: split is very useful for handling huge files, backups, or transfers over limited bandwidth.

181. How do you merge files created with the split command?

Answer: Use the cat command:

```
$ cat part_* > largefile.txt
```

Example Output

```
$ cat part_* > largefile.txt
$ ls -lh largefile.txt
-rw-r--r-- 1 user user 327M May 28 10:15 largefile.txt
```

NOTE: Make sure the parts are in the same directory and named in the correct order (aa, ab, ac...).

182. How do you replace text in a file from the command line?

Answer: Use the sed command:

```
$ sed 's/old_text/new_text/g' filename
```

Example Output

```
$ cat before.txt
Linux is old_text and old_text.
$ sed 's/old_text/new_text/g' before.txt > after.txt
$ cat after.txt
Linux is new_text and new_text.
```

TIP: The 'g' flag replaces all occurrences in a line. Remove 'g' to replace only the first match.

183. How do you print a specific column from a text file?

Answer: Use the awk command:

```
$ awk '{print $2}' filename
```

Example Output

```
$ cat users.txt
alice devops  1001
bob   sysadmin 1002
charlie cloud  1003
david security 1004
```

→

```
$ awk '{print $2}' users.txt
devops
sysadmin
cloud
security
```

BEST PRACTICE: awk is extremely powerful for text processing and report generation.

184. How do you count the number of matching lines in a file?

Answer: Use the grep -c command:

```
$ grep -c "pattern" filename
```

Example Output

```
$ cat app.log
INFO  Starting service
ERROR  Failed to connect
INFO   Retrying connection
ERROR  Timeout occurred
INFO   Service restarted
```

→

```
$ grep -c "ERROR" app.log
2
```

NOTE: -c option shows only the count of matching lines.

185. How do you display only the unique lines in a file?

Answer: Use the sort -u command:

```
$ sort -u filename
```

Example Output

```
$ cat items.txt
apple
banana
apple
orange
banana
grape
```

→

```
$ sort -u items.txt
apple
banana
grape
orange
```

TIP: sort -u first sorts the file, then removes duplicate lines.

186. How do you display line numbers while viewing a file?

Answer: Use the nl command:

```
$ nl filename
```

Example Output

```
$ cat config.txt
server=localhost
port=8080
user=admin
debug=true
```

→

```
$ nl config.txt
   1    server=localhost
   2    port=8080
   3    user=admin
   4    debug=true
```

BEST PRACTICE: nl is helpful for debugging scripts and configurations.

PRODUCTION INSIGHT: These commands help you handle, analyze, and transform data efficiently. Mastering them is essential for automation, DevOps, and troubleshooting.

AUTOMATE. ANALYZE. SIMPLIFY. DELIVER.

200 LINUX INTERVIEW Q&A

187. How do you display the beginning of every line in a file with line numbers?

Answer: Use the cat -n command:

```
$ cat -n filename
```

Example Output (cat -n filename)

```
$ cat -n users.txt
  1  alice
  2  bob
  3  charlie
  4  david
  5  eve
```

TIP: cat -n is very useful when referencing specific lines in configuration files or scripts.

188. How do you continuously monitor new entries added to a log file?

Answer: Use the tail -f command:

```
$ tail -f /var/log/syslog
```

Example Output (tail -f /var/log/syslog)

```
$ tail -f /var/log/syslog
May 28 10:21:01 server sshd[1234]: Accepted password for user root
May 28 10:21:05 server CRON[1250]: (root) CMD (   cd / && run-parts
--report /etc/cron.hourly)
May 28 10:21:07 server systemd[1]: Started Session 123 of user root.
May 28 10:21:10 server sshd[1280]: Failed password for invalid user test
```

NOTE: tail -f is essential for real-time log monitoring and troubleshooting production issues.

189. How do you monitor a command's output continuously at regular intervals?

Answer: Use the watch command:

```
$ watch df -h
```

Example Output (watch df -h)

```
Every 2.0s: df -h                server: Wed May 28 10:22:30 2025

Filesystem      Size  Used Avail Use% Mounted on
udev            3.9G     0  3.9G   0% /dev
tmpfs           795M  1.6M  794M   1% /run
/dev/sda1        50G   12G   36G  25% /
tmpfs           3.9G     0  3.9G   0% /dev/shm
```

TIP: wail -f is great for observing changing system metrics, disk usage, and performance.

190. How do you execute a command with elevated priority?

Answer: Use the nice command:

```
$ nice -n -10 command_name
```

Example Output (nice -n -10 command_name)

```
$ nice -n -10 /usr/bin/backup.sh
[1] 2356
$ ps -o pid,ni,comm -p 2356
   PID  NI COMMAND
  2356 -10 backup.sh
```

BEST PRACTICE: Use higher priority (negative nice value) only for critical system tasks.

191. How do you change the priority of a running process?

Answer: Use the renice command:

```
$ sudo renice -5 -p PID
```

Example Output (sudo renice -5 -p 2356)

```
$ ps -o pid,ni,comm -p 2356
  PID  NI COMMAND
 2356   0 backup.sh
$ sudo renice -5 -p 2356
2356 (process ID) old priority 0, new priority -5
$ ps -o pid,ni,comm -p 2356
  PID  NI COMMAND
 2356  -5 backup.sh
```

NOTE: Lower nice value (negative) = higher priority. Requires root privileges.

192. How do you display a process's scheduling priority?

Answer: Use the ps command with the ni (nice value) column:

```
$ ps -o pid,ni,comm
```

Example Output (ps -o pid,ni,comm)

```
  PID  NI COMMAND
    1   0 systemd
  432 -10 backup.sh
  987   0 sshd
 1234   5 python3
```

TIP: The NI column shows the nice value of each process. Range: -20 (highest priority) to 19 (lowest priority).

193. How do you execute a command as another user without switching accounts?

Answer: Use the sudo -u command:

```
$ sudo -u username command_name
```

Example Output (sudo -u www-data ls -l /var/www)

```
$ sudo -u www-data ls -l /var/www
total 12
-rw-r--r-- 1 www-data www-data  1234 May 28 10:00 index.html
drwxr-xr-x 2 www-data www-data  4096 May 28 10:00 images
-rw-r--r-- 1 www-data www-data   567 May 28 10:00 style.css
```

BEST PRACTICE: sudo -u is useful for testing permissions or running tasks as a specific user.

PRODUCTION INSIGHT: Knowing process priority and monitoring commands helps you maintain system performance, troubleshoot issues effectively, and ensure smooth operations in production environments.

MONITOR. PRIORITIZE. OPTIMIZE. DELIVER.

200 LINUX INTERVIEW Q&A

194. How do you execute a command in the background?

Answer: Add an ampersand (&) at the end of the command:

```
$ long_running_command &
```

Example Output

```
$ sleep 1000 &
[1] 4321
$ jobs
[1]+  Running                 sleep 1000 &
$
```

TIP: The ampersand (&) lets the command run in the background so you can continue using the shell.

195. How do you display all background jobs in the current shell?

Answer: Use the jobs command:

```
$ jobs
```

Example Output

```
$ jobs
[1]   Running                 sleep 1000 &
[2]-  Running                 tar -czf backup.tar.gz /data &
[3]+  Stopped                 vim notes.txt
$
```

NOTE: The + indicates the most recent job. The - indicates the previous job.

196. How do you bring a background job to the foreground?

Answer: Use the fg command:

```
$ fg %1
```

Example Output

```
$ jobs
[1]   Running                 sleep 1000 &
[2]-  Running                 tar -czf backup.tar.gz /data &
$ fg %1
sleep 1000
(sleep runs in the foreground)
```

TIP: Replace %1 with the job number you want to bring to the foreground.

197. How do you move a running foreground process to the background?

Answer: Press Ctrl + Z to suspend the process, then use the bg command:

```
$ bg %1
```

Example Output

```
$ long_running_command
^Z
[1]+  Stopped                 long_running_command
$ bg %1
[1]+  long_running_command &
$ jobs
[1]+  Running                 long_running_command &
$
```

NOTE: Ctrl + Z suspends the process and returns you to the shell prompt.

198. How do you keep a process running after logging out?

Answer: Use the nohup command:

```
$ nohup command_name &
```

Example Output

```
$ nohup long_task.sh > output.log 2>&1 &
[1] 5678
$ nohup: ignoring input and appending output to 'nohup.out'
$
```

TIP: nohup ensures the process ignores hangups (SIGHUP) when you log out.

199. How do you display the absolute path of the current working directory?

Answer: Use the pwd -P command:

```
$ pwd -P
```

Example Output

```
$ pwd
./projects/linux
$ pwd -P
/home/user/projects/linux
$
```

BEST PRACTICE: pwd -P resolves symbolic links and shows the true physical path.

200. How do you execute a command at a specific date and time using systemd?

Answer: Use the systemd-run command with the --on-calendar option:

```
$ systemd-run --on-calendar="2026-08-01 10:00:00" \
    command_name
```

Example Output

```
$ systemd-run --on-calendar="2026-08-01 10:00:00" \
        /usr/local/bin/backup.sh
Running as unit: run-r6b7c8d9e0f1412bb3e5.service

$ systemctl list-timers --all | grep backup
Sat 2026-08-01 10:00:00 UTC   2 months left   run-r6b7c8d9e0f1412bb3e5.service
$
```

NOTE: The command will run only once at the scheduled date and time unless a timer unit is created.

PRODUCTION INSIGHT: Mastering job control, process management, and scheduling commands is essential for running reliable services, automations, and long-running tasks in production environments.

AUTOMATE. SCHEDULE. STAY IN CONTROL.

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Linux File Permissions Cheat Sheet

1. PERMISSION MODEL (r, w, x)

r	w	x
Read	Write	Execute

Owner			Group			Others		
r	w	x	r	w	x	r	w	x
1	2	4	1	2	4	1	2	4

Each position has a value:
r = 4 w = 2 x = 1
Add the values to get the number.

2. NUMERIC PERMISSIONS

Permission	Octal	Meaning
400		Read only (r--------)
600		Read + Write (rw------)
644		Owner RW, Group R, Others R (rw-r--r--)
700		Owner Full Access (rwx------)
755		Owner Full, Group RX, Others RX (rwxr-xr-x)
777		Everyone Full Access (rwxrwxrwx) DANGER: Never use in production!

3. SYMBOLIC PERMISSIONS

```
chmod u+x script.sh      # Add execute for owner
chmod g-w file.txt       # Remove write for group
chmod o-r secret.txt     # Remove read for others
chmod a+r file.txt       # Add read for all (u,g,o)
chmod u=rw,g=r,o=        # Set exact permissions
chmod -R u+rwX folder    # Recursive change
```

u = user (owner) g = group
o = others a = all (u,g,o)
+ = add - = remove = = set exact

4. CHMOD QUICK REFERENCE

Command	Result
chmod 644 file.txt	rw-r--r--
chmod 755 script.sh	rwxr-xr-x
chmod 600 private.key	rw-------
chmod -R 755 folder/	Recursively set 755
chmod u+x file.sh	Add execute for owner
chmod go-w file.txt	Remove write for group & others

```
$ ls -l file.txt
-rw-r--r-- 1 user staff  1024 May 28 10:00 file.txt
```

5. CHOWN vs CHGRP

chown	chgrp
Changes the owner of a file/directory. Can also change the group.	Changes only the group of a file/directory.

Examples

```
sudo chown user file
sudo chown user:group file
sudo chown -R user:group dir/
sudo chgrp developers file
sudo chgrp -R developers dir/
```

NOTE:
Use chown to change ownership.
Use chgrp to adjust group without affecting ownership.

6. DEFAULT UMASK VALUES

umask	Files	Directories	Meaning
022	644	755	Standard default
027	640	750	More restrictive
077	600	700	Highly secure

```
$ umask
0022    ← Current umask
$ umask 027  # Set for current shell session
```

umask removes permissions by default from newly created files and directories.
It is SUBTRACTED from 777.
Example: 777 - 022 = 755 (directories)
777 - 022 = 644 (files)

7. SPECIAL PERMISSIONS

Permission	Description	chmod Example	Shown As	Common Use
SUID	Runs a file with the privileges of the file owner.	chmod u+s file	rwsr-xr-x	/usr/bin/passwd
SGID	Runs with the privileges of the file group.	chmod g+s file	rwxr-sr-x	Shared directories
Sticky Bit	Users can delete only their own files in the directory.	chmod +t directory	rwxrwxrwt	/tmp

Check with: ls -l s (SUID/SGID) replaces x t (Sticky) replaces x in others position

8. PRODUCTION PERMISSION EXAMPLES

Web server files (HTML/CSS/JS)	755 (rwxr-xr-x)
SSH private key	600 (rw-------)
Executable scripts	755 (rwxr-xr-x)
Configuration files	640 (rw-r-----)
Log files	640 (rw-r-----)
Shared directories (SGID)	2775 (rwxrwsr-x)

9. SECURITY REMINDERS

- [x] Apply least privilege principle.
- [x] Avoid 777 permissions.
- [x] Audit file permissions regularly.
- [x] Protect SSH keys with 600.
- [x] Verify ownership after deployments.
- [x] Use groups instead of broad permissions.
- [x] Monitor sensitive files and directories.
- [x] Review umask and special permissions.

QUICK CHECK COMMANDS

```
ls -l                # List permissions
stat file            # Detailed info
namei -l /path       # Path permissions
find / -perm 777 -type f 2>/dev/null
     # Find dangerous permissions
```

PRODUCTION INSIGHT

Incorrect file permissions are a common cause of security incidents and deployment failures. Always grant the minimum permissions required for the workload.

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Linux File System Hierarchy Cheat Sheet

DIRECTORY	PURPOSE	KEY CONTENT
/	Root of the entire file system	Top-level directory
/bin	Essential user commands	ls, cp, mv, cat, etc.
/sbin	System administration commands	ifconfig, reboot, fsck, etc.
/etc	System configuration files	Configs for services, users, host
/usr	User programs and data	Applications, libraries, docs
/var	Variable data files (logs, cache)	Logs, spool, temp, databases
/home	User home directories	User data and settings
/root	Root user home directory	Superuser (high privilege) data
/tmp	Temporary files	Short-lived files, system temp
/opt	Optional third-party software	Custom applications, packages
/boot	Boot loader and kernel files	Kernel, initrd, bootloader files
/dev	Device files (hardware)	Disk, USB, tty, null, etc.
/proc	Process and kernel information	Virtual files for running system
/sys	System and hardware info	Kernel and hardware interfaces
/run	Runtime data (volatile)	PID files, sockets, system state

COMMON PRODUCTION LOCATIONS

- /var/log – Application and system logs
- /etc/nginx – Web server configuration
- /etc/systemd/system – Service units
- /var/www – Web content (web servers)
- /opt – Third-party applications
- /home – User data and configurations
- /root – Root user files and configs
- /tmp – Temporary files (auto-cleaned)
- /backup – Backup and data archives

WHERE ENGINEERS TROUBLESHOOT

- /var/log – Check logs for errors
- /etc – Verify configurations
- /proc – Check running processes
- /sys – Inspect hardware and kernel
- /var/tmp – Check temporary issues
- /home – User-related problems
- /dev – Device and storage issues
- /run – Service and runtime problems

KEY TAKEAWAYS

- Understand where files live.
- Know what each directory is for.
- Troubleshoot by checking the right locations first.
- Use commands like ls, du, df, cat, less, and journalctl.

QUICK COMMANDS

`ls /` – List root directory

`du -sh /var` – Size of /var

`df -h` – Disk usage

`find / -type f -name "*.log"` – Find logs

`grep "error" /var/log/syslog` – Search logs

EXAMPLE PRACTICE

- Logs? → /var/log/
- Config files? → /etc/
- User data? → /home/
- Running processes? → /proc, /run
- Device issues? → /dev
- System info? → /sys

TROUBLESHOOTING TIPS

ISSUE	CHECK HERE
Service not running	/etc/systemd/system, /var/log
High disk usage	/var, /tmp, /home, /var/log
Permissions problems	/etc/passwd, /etc/group, file owners
Network issues	/etc/network, /var/log, /proc/net
System boot issues	/boot, /etc/fstab, dmesg, journalctl
Hardware issues	/dev, /sys, dmesg

SECURITY REMINDERS

- Keep /etc secure (chmod, chown).
- Do not store sensitive data in /tmp.
- Monitor logs in /var/log regularly.
- Review permissions on critical files.
- Use least privilege principle.
- Backups and monitoring save systems.

Know your filesystem. Protect your system. Troubleshoot faster. Stay secure.

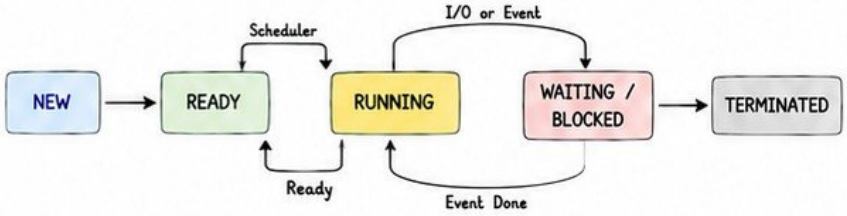
Good knowledge of the filesystem = Better Troubleshooting + Stronger Systems

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Linux Process Management Cheat Sheet

1. PROCESS LIFECYCLE



2. PROCESS STATES

STATE	DESCRIPTION
New	Process is being created.
Ready	Waiting in the ready queue for CPU.
Running	Currently using the CPU.
Waiting / Blocked	Waiting for I/O or event to complete.
Terminated	Process has finished execution.

3. SIGNALS OVERVIEW

Signals are used to communicate with processes. They can be sent from the kernel, the terminal, or other processes.

4. COMMON SIGNAL NUMBERS

SIGNAL	NUMBER	DEFAULT ACTION	DESCRIPTION
SIGHUP	1	Terminate	Hangup detected on controlling terminal
SIGINT	2	Terminate	Interrupt from keyboard (Ctrl + C)
SIGQUIT	3	Core Dump	Quit from keyboard (Ctrl + \)
SIGKILL	9	Terminate	Kill signal (cannot be caught or ignored)
SIGTERM	15	Terminate	Terminate signal (polite request)
SIGSTOP	19	Stop	Stop process (cannot be caught or ignored)
SIGCONT	18	Continue	Continue a stopped process
SIGCHLD	17	Ignore	Child process stopped or terminated

5. KILL WORKFLOW

1. Check the process
   `ps -ef | grep process`
2. Send SIGTERM (15)
   `kill -15 PID`
3. Wait a few seconds
   5-10 sec
4. If not stopped, send SIGKILL (9)
   `kill -9 PID`
5. Verify it's gone
   `ps -p PID`

Best Practice:
Always try SIGTERM (15) before SIGKILL (9). SIGKILL should be a last resort.

6. JOB CONTROL

Command	Description
Ctrl + Z	Suspend foreground job
bg %job	Resume job in background
fg %job	Bring job to foreground
jobs	List background jobs
kill %job	Send signal to a job

Job ID format: %1, %2, %3 ...
Example:

```
$ jobs
[1]+  Stopped                 vim file.txt
[2]-  Running                 sleep 1000 &
```

7. BACKGROUND vs FOREGROUND

FOREGROUND
- Runs in the current terminal.
- You interact with it directly.
- Terminal is blocked until it finishes.

`$ command_name`

BACKGROUND
- Runs independently.
- Terminal is free for other commands.
- Add & at the end.

`$ command_name &`

8. NICE VALUES



- Range: -20 (highest priority) to +19 (lowest)
- Lower values = higher CPU priority
- Only root can use negative values.

9. RENICE EXAMPLES

```
# View nice values
$ ps -o pid,ni,comm
 PID    NI   COMMAND
 1234    0   nginx
 2345   10   backup.sh

# Change nice value
$ sudo renice -5 -p 2345
 2345 (process ID) old priority 10, new priority -5

# Change for multiple processes
$ sudo renice 5 -p 1234,2345,3456
```

10. ZOMBIE PROCESS NOTES

A zombie process is a terminated process that still has an entry in the process table because its parent has not read its exit status.

- Zombies do not consume CPU or memory.
- They only consume a PID.
- Too many zombies can exhaust the PID table.
- Fix: Restart or kill the parent process.

```
# Find zombie processes
$ ps -eo pid,ppid,state,cmd | grep '^Z'
1256     1      Z      [defunct]
```

11. PRODUCTION TROUBLESHOOTING WORKFLOW

1. **Identify**
   top, htop
   ps, pstree
   pgrep, pidof
2. **Analyze**
   CPU, MEM, I/O
   Threads
   Open files
   Network
3. **Act**
   Adjust priority
   Kill or stop
   Restart service
4. **Verify**
   Monitor again
   Confirm fix
   Check logs
5. **Prevent**
   Set limits
   Tuning
   Monitoring
   Alerts

Useful Commands: `top` | `htop` | `ps aux` | `pstree` | `pmap` | `lsof -p PID` | `strace -p PID` | `dmesg` | `journalctl`

Remember:
Understand your processes. Monitor them. Control them. Optimize them.
Great systems are built by engineers who master process management.

KEY TAKEAWAY
A healthy system is the result of well-managed, well-understood processes.

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Linux Networking Cheat Sheet

1. IP COMMANDS

Purpose	Command
Show all IP addresses	ip addr show
Show IPv4 addresses	ip -4 addr show
Show IPv6 addresses	ip -6 addr show
Add IP address	ip addr add 10.0.0.10/24 dev eth0
Delete IP address	ip addr del 10.0.0.10/24 dev eth0
Bring interface up	ip link set eth0 up
Bring interface down	ip link set eth0 down
Show link status	ip link show

2. ROUTE COMMANDS

Purpose	Command
Show routing table	ip route show
Show default route	ip route show default
Add route	ip route add 10.0.0.0/24 via 10.0.0.1
Delete route	ip route del 10.0.0.0/24
Add default route	ip route add default via 10.0.0.1
Show kernel routes	ip -s route show

Routes tell the kernel where to send packets. The default route is your exit to other networks.

3. DNS COMMANDS

Purpose	Command
Test DNS resolution	nslookup google.com
Another DNS tool	dig google.com
Short output	dig +short google.com
Reverse lookup	dig -x 8.8.8.8
Check DNS servers	cat /etc/resolv.conf
Flush DNS cache (systemd-resolved)	resolvectl flush-caches

4. HOSTNAME COMMANDS

Purpose	Command
Show hostname	hostname
Show FQDN	hostname -f
Set hostname (temp)	hostname newname
Set hostname (persistent)	sudo hostnamectl set-hostname newname
Show DNS domain	hostname -d
Show all details	hostnamectl

Persistent hostname change updates /etc/hostname and /etc/hosts automatically.

5. INTERFACE COMMANDS

Purpose	Command
Show all interfaces	ip link show
Show interfaces (legacy)	ifconfig -a
Interface statistics	ip -s link show eth0
Enable interface	ip link set eth0 up
Disable interface	ip link set eth0 down
View ARP table	ip neigh show
Add static ARP entry	ip neigh add 10.0.0.20 lladdr aa:bb:cc:dd:ee:ff dev eth0

6. SOCKET COMMANDS

Purpose	Command
Show listening sockets	ss -tuln
Show all sockets	ss -a
Show TCP sockets	ss -t
Show UDP sockets	ss -u
Show process using port	ss -tuhp
Show summary	ss -s

ss is faster and more powerful than netstat.

7. PORT INSPECTION

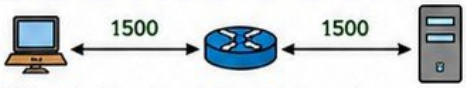
Purpose	Command
Check if port open (TCP)	nc -vz 192.168.1.10 80
Check if port open (UDP)	nc -zvu 192.168.1.10 53
List open ports (listening)	sudo lsof -i -P -n
List open ports by process	sudo ss -tulnp
Check specific port	sudo ss -tulnp | grep :80

Common ports: 22 SSH, 53 DNS, 80 HTTP, 443 HTTPS, 3306 MySQL, 5432 PostgreSQL

8. CONNECTIVITY TESTING

Purpose	Command
Test basic connectivity	ping -c 4 8.8.8.8
Ping with interface	ping -I eth0 8.8.8.8
Trace route	traceroute google.com
Trace route with ICMP	traceroute -I google.com
MTR (better traceroute)	mtr google.com
Check HTTP/HTTPS	curl -I https://google.com
Check DNS + HTTP	curl -I https://google.com

9. MTU (MAX TRANSMISSION UNIT)



MTU is the largest packet size that can be transmitted without fragmentation.

Check MTU:

```
$ ip link show eth0
```

Set MTU (temporary):

```
$ sudo ip link set eth0 mtu 1400
```

10. DEFAULT GATEWAY

The default gateway is the router used when the destination is outside your local network.

Show default gateway:

```
$ ip route show default
```

Add default gateway:

```
$ sudo ip route add default via 192.168.1.1
```

Remove default gateway:

```
$ sudo ip route del default via 192.168.1.1
```

11. CIDR QUICK REFERENCE

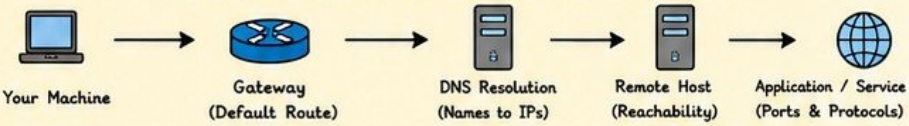
CIDR	Netmask	Hosts	CIDR	Netmask	Hosts
/32	255.255.255.255	1	/26	255.255.255.192	62
/31	255.255.255.254	2	/25	255.255.255.128	126
/30	255.255.255.252	2	/24	255.255.255. 0	254
/29	255.255.255.248	6	/23	255.255.254. 0	510
/28	255.255.255.240	14	/22	255.255.252. 0	1022
/27	255.255.255.224	30	/21	255.255.248. 0	2046

Hosts = Usable IPs (Network + Broadcast excluded)

12. NETWORK TROUBLESHOOTING WORKFLOW

1. Identify the problem
   What is not working?
2. Check local configuration
   IP, route, DNS, interface status
3. Test connectivity step by step
   ping → gateway → remote host
4. Inspect routes & firewall
   ip route, firewall rules
5. Check logs & services
   system logs, DNS, DHCP, services
6. Fix, verify, and monitor
   Confirm resolution & monitor

NETWORK TROUBLESHOOTING ORDER (ALWAYS FOLLOW)



BEST PRACTICES

- Keep interfaces up and secure.
- Use static IPs for critical systems.
- Monitor latency, packet loss, and errors.
- Document network changes.
- Test from multiple points (inside & outside).

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Linux Storage Cheat Sheet

1. lsblk – List Block Devices

Displays block devices and their mount points in a tree-like format.

```
$ lsblk -f
NAME        FSTYPE      SIZE    MOUNTPOINT
sda                     100G
├─sda1      ext4          1G    /boot
├─sda2      LVM2_member  99G
│ ├─vg-root ext4         50G    /
│ └─vg-home ext4         49G    /home
sdb         ext4        200G    /data
```

Useful options:

- `lsblk` # Basic info
- `lsblk -f` # Show filesystem, UUID
- `lsblk -o NAME,SIZE,TYPE,MOUNTPOINT`
- `lsblk -e 7` # Exclude loop devices

2. blkid – Show Block Device UUIDs

Displays UUID and filesystem type of devices.

```
$ sudo blkid
/dev/sda1: UUID="a1b2c344-1111-2222-3333-aaaaaaaabbbb"
           TYPE="ext4" PARTUUID="0001"
/dev/sda2: UUID="b2c3d455-3333-4444-5555-bbbbbbbbbbbb"
           TYPE="LVM2_member"
/dev/sdb1: UUID="c3d4e5f6-5555-6666-7777-cccccccccc"
           TYPE="ext4"
```

Tip:
UUIDs do not change even if the device name changes. Use UUIDs in /etc/fstab for reliability.

3. df – Disk Free (File Systems)

Shows disk space usage of mounted filesystems.

```
$ df -h
Filesystem      Size  Used Avail Use% Mounted on
/dev/sda1       1.0G  240M  760M  24% /boot
/dev/mapper/vg-root  50G   12G   36G  25% /
/dev/mapper/vg-home  49G  5.6G   41G  12% /home
/dev/sdb1       200G   30G  160G  16% /data
tmpfs           2.0G     0  2.0G   0% /run
```

Useful options:

- `df -h` # Human readable
- `df -T` # Show filesystem type
- `df -i` # Show inode usage

4. du – Disk Usage (Directories)

Shows size of files and directories.

```
$ du -sh /var/log
1.2G    /var/log
```

Useful options:

- `du -sh *` # Summary of items
- `du -ah /path | less` # Detailed list
- `du --max-depth=1 /path` # Limit directory depth
- `du -h` # Human readable

5. mount – Mount Filesystems

Mounts a filesystem to a directory.

```
$ sudo mount /dev/sdb1 /data
$ mount | grep /data
/dev/sdb1 on /data type ext4 (rw,relatime)
```

Useful options:

- `mount` # Show mounted filesystems
- `mount -t ext4 /dev/sdb1 /data`
- `mount -a` # Mount all from fstab
- `mount -o ro /dev/sdb1 /mnt` # Read-only

6. umount – Unmount Filesystems

Unmounts a mounted filesystem.

```
$ sudo umount /data
$ umount -l /data    # Lazy umount (if busy)
```

Warning:
Ensure no process is using the mount point before unmounting.

7. fstab – File System Table

Defines filesystems to mount at boot.

```
# <file system>                <mount point> <type>  <options> <dump> <pass>
UUID=a1b2c344-1111-2222-3333-aaaaaaaabbbb  /boot   defaults  1      2
UUID=b2c3d455-3333-4444-5555-bbbbbbbbbbbb  /      defaults  1      1
UUID=c3d4e5f6-5555-6666-7777-cccccccccccc  /data  defaults  0      2
/swapfile                                  none   swap      0      0
```

Useful commands:

- `sudo mount -a` # Mount all from fstab
- `sudo fsck -av` # Check filesystems in fstab
- `man fstab` # View fstab manual

8. UUID – Universally Unique Identifier

Unique ID for block devices and filesystems.

```
$ sudo blkid /dev/sdb1
/dev/sdb1: UUID="c3d4e5f6-5555-6666-
7777-cccccccc" TYPE="ext4"
PARTUUID="12345678-01"
```

Why use UUID in /etc/fstab?

- Stable across reboots
- Not affected by device name changes
- Prevents mount failures

9. Swap – Memory Swap Space

Swap extends virtual memory to disk.

```
$ free -h
Mem:    total  used  free  shared buff/cache  avail
        7.7G   3.2G  1.1G   300M     3.4G     4.0G
Swap:   2.0G   256M  1.7G
```

Create swap file:

1. `sudo fallocate -l 2G /swapfile`
2. `sudo chmod 600 /swapfile`
3. `sudo mkswap /swapfile`
4. `sudo swapon /swapfile`

Make permanent: Add to /etc/fstab

```
/swapfile  none  swap  sw  0  0
```

10. File Systems – Overview

Filesystem	Features	Best Use Cases
ext4	Stable, journaling, large volumes	General purpose (Linux default)
xfs	High performance, scalability	Databases, large filesystems
btrfs	Snapshots, compression, RAID, checksums	Advanced storage, snapshots
swap	Virtual memory extension	System swap
vfat	FAT32 compatible	USB drives, EFI partitions
ntfs	Windows compatibility	Shared with Windows
tmpfs	In-memory filesystem	Temporary files, RAM disk

11. Common Mount Options

Option	Description
defaults	rw, suid, dev, exec, auto, nouser, async
ro	Mount read-only
rw	Mount read-write
noexec	Do not allow execution of binaries
nosuid	Ignore SUID and SGID bits
nodev	Do not interpret character or block devices
noatime	Do not update access time
relatime	Update access time relative to modify time
discard	Enable TRIM (SSD)
nfsvers=4	Specify NFS version

Quick Examples

- Read-only mount:
  `$ sudo mount -o ro /dev/sdb1 /mnt`
- Mount all from fstab:
  `$ sudo mount -a`
- Remount as read-write:
  `$ sudo mount -o remount,rw /mnt`
- Check mount type:
  `$ mount | grep /data`
- Show disk usage in human format:
  `$ df -h`
- Show large directories:
  `$ du -sh /* | sort -h`

12. STORAGE TROUBLESHOOTING CHECKLIST

- ☐ Check disk space: `df -h`
- ☐ Check inode usage: `df -i`
- ☐ Check large directories: `du -sh /* | sort -h`
- ☐ Check mounted devices: `lsblk -f`
- ☐ Verify mounts: `mount | column -t`
- ☐ Check fstab syntax: `sudo mount -a`
- ☐ Check filesystem health: `sudo fsck -f /dev/sdX`
- ☐ Check SMART status: `sudo smartctl -a /dev/sdX`
- ☐ Check I/O errors: `dmesg | grep -i error`
- ☐ Check swap usage: `free -h`
- ☐ Review logs: `dmesg | less /var/log/syslog`

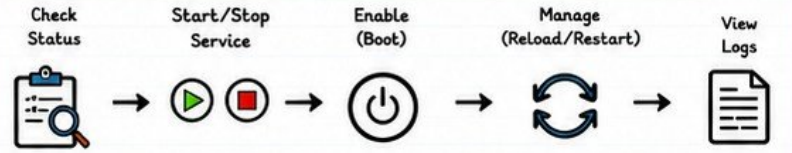
Remember:
Good storage hygiene = Stable, high-performing systems.

- Always backup before making storage changes.
- Use UUIDs in /etc/fstab for reliability.
- Monitor disk health and capacity regularly.

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Linux Service Management Cheat Sheet

1. systemctl WORKFLOW



2. SERVICE LIFECYCLE (STATE)

State	Description
loaded	Unit file is loaded by systemd.
active (running)	Service is running.
active (exited)	Service ran and exited successfully.
inactive (dead)	Service is stopped.
failed	Service failed to start.
activating	Service is in the process of starting.
deactivating	Service is in the process of stopping.

3. ENABLE (START ON BOOT)

Starts the service now (if not running) and enables it to start at boot.

```
$ sudo systemctl enable nginx
$ sudo systemctl enable --now nginx
$ sudo systemctl enable nginx
# (enable only, do not start)
```

Check:
systemctl is-enabled nginx
systemctl status nginx

4. DISABLE (STOP START ON BOOT)

Disables the service from starting automatically at boot.

```
$ sudo systemctl disable nginx
$ sudo systemctl disable --now nginx
```

Check:
systemctl is-enabled nginx
systemctl status nginx

5. MASK (PREVENT START)

Prevents a service from being started manually or automatically.

```
$ sudo systemctl mask nginx
```

Check:
systemctl is-enabled nginx
should show: masked

6. UNMASK (ALLOW START)

Removes the mask and allows the service to be started.

```
$ sudo systemctl unmask nginx
```

Check:
systemctl is-enabled nginx
should not show: masked

7. RELOAD (RE-READ CONFIG)

Reloads the service configuration without stopping the service.

```
$ sudo systemctl reload nginx
```

Use when:
- Service supports reload.
- You changed config files.
- You want zero downtime.

Check:
sudo systemctl status nginx

8. RESTART (BOUNCE SERVICE)

Stops and starts the service. Use when reload is not supported or service not behaving correctly.

```
$ sudo systemctl restart nginx
```

Check:
sudo systemctl status nginx
journalctl -u nginx --no-pager

STATUS (SERVICE STATUS)

Shows detailed status, health, and recent log lines.

```
$ sudo systemctl status nginx
```

Look for:
- Active: active (running)
- Main PID
- Memory/CPU
- Recent log messages

10. JOURNAL INTEGRATION

View and filter logs for a service using journalctl.

```
$ journalctl -u nginx --no-pager
$ journalctl -u nginx -f
$ journalctl -u nginx -n 100
$ journalctl -u nginx --since "1 hour ago"
```

# -f	Follow logs (live)
# -n 100	Last 100 lines
# --since	Show logs since time

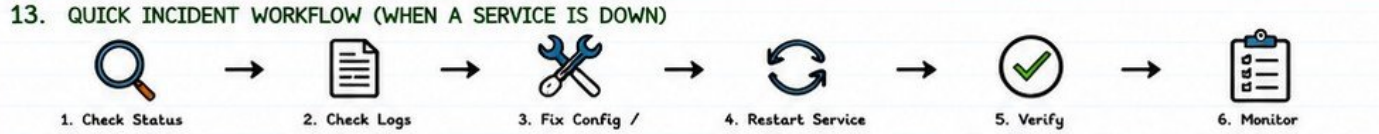
11. COMMON SERVICES (NAMES MAY VARY BY DISTRIBUTION)

Service	Description	Common Unit Name
SSH	Secure Shell server	sshd
NGINX	Web server / Reverse proxy	nginx
Apache	Web server	httpd / apache2
MySQL	MySQL database server	mysqld
PostgreSQL	PostgreSQL database server	postgresql
Docker	Docker daemon	docker
Network	Network management	NetworkManager
Firewall	Firewall service	firewalld / ufw
Cron	Task scheduler	crond / cron

12. COMMON SERVICE TROUBLESHOOTING

Problem	Possible Causes	What to Check / Fix
Service not starting	Bad config, port in use, permissions, dependencies	systemctl status <svc> journalctl -u <svc> --no-pager Check config syntax
Start then stop	Config error, missing dependency	journalctl -u <svc> -n 100 Fix config / missing files
Port already in use	Another service using same port	ss -tulnp | grep :80 Change port or stop other svc
Service slow	High load, I/O, network, resource limits	top / htop iotop / iftop Check logs
High memory usage	Leak, configuration, large workload	systemctl status <svc> Check memory usage over time
Service crashes	Bugs, OOM, wrong config, missing files	journalctl -u <svc> -xe Check config and dependencies

13. QUICK INCIDENT WORKFLOW (WHEN A SERVICE IS DOWN)



BEST PRACTICES

- Prefer reload over restart for zero downtime.
- Always check status and logs before making changes.
- Enable only services that must start at boot.
- Use systemctl cat <service> to view unit file.
- Keep configs validated before reloading/restarting.

REMEMBER

- systemctl is the modern way to manage services.
- Use journalctl for deep log investigation.
- A healthy service = stable system.
- Logs tell you WHY, status tells you WHAT.

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Linux Log Analysis Cheat Sheet

1. journalctl – Systemd Logs

Query and analyze systemd journal logs.

```
$ journalctl -xe
$ journalctl -u nginx
$ journalctl -u nginx --since "1 hour ago"
$ journalctl -u nginx --until "10 min ago"
$ journalctl -p err -b
$ journalctl --boot=-1
```

Options:

-u, --unit	Filter by service unit
-b, --boot	Filter by boot ID
-p, --priority	Filter by log level
--since	Show logs since time
--until	Show logs until time
-xe	Jump to end and show details

2. dmesg – Kernel Ring Buffer

View kernel messages from the buffer.

```
$ dmesg
$ dmesg -T
$ dmesg | less
$ dmesg -w
```

Use for:

- Hardware errors
- Driver issues
- Boot problems
- Kernel panics



3. syslog – System Logs

General system log messages.

```
$ less /var/log/syslog
$ tail -f /var/log/syslog
$ grep "error" /var/log/syslog
```

Contains:

- System events
- Service messages
- Daemon logs
- Cron jobs
- User sessions



4. auth.log – Auth Logs

Authentication and authorization logs.

```
$ less /var/log/auth.log
$ grep "Failed" /var/log/auth.log
$ grep "sshd" /var/log/auth.log
```

Use for:

- SSH logins
- Sudo usage
- PAM events
- Failed logins
- Privilege escalation



5. kern.log – Kernel Logs

Kernel related log messages.

```
$ less /var/log/kern.log
$ grep -i "error" /var/log/kern.log
$ grep -i "fail" /var/log/kern.log
```

Use for:

- Kernel errors
- Hardware failures
- Driver issues
- OOM events
- System crashes

6. tail – Real-time Log Monitoring

Display last part of files.

```
$ tail /var/log/syslog
$ tail -n 100 /var/log/auth.log
$ tail -f /var/log/nginx/access.log
$ tail -F /var/log/app.log
```

Options:

-f	Follow (real-time)
-F	Follow by name (reopen if rotated)
-n	Show last N lines



7. less – View Large Log Files

View and navigate large log files.

```
$ less /var/log/syslog
$ less /var/log/dmesg
$ less +F /var/log/auth.log
```

Keys:

/pattern	Search forward
?pattern	Search backward
n	Next match
N	Previous match
q	Quit



8. grep – Search Patterns

Search for patterns in logs.

```
$ grep "error" /var/log/syslog
$ grep -i "failed" /var/log/auth.log
$ grep -E "error|fail|denied" /var/log/syslog
$ grep -n "timeout" /var/log/syslog
```

Options:

-i	Ignore case
-E	Extended regex
-n	Show line numbers
-r	Recursive search



9. awk – Pattern Processing

Extract and analyze log fields.

```
$ awk '{print $1,$2,$3}' /var/log/syslog
$ awk '$5=="error"' /var/log/syslog
$ awk -F' ' '{print $1,$NF}'
    /var/log/auth.log
```

Use for:

- Field extraction
- Summarization
- Counting
- Pattern matching



10. sed – Stream Editor

Modify and filter log lines.

```
$ sed -n '1,100p' /var/log/syslog
$ sed 's/failed/FAILED/g' /var/log/auth.log
$ sed '/error/d' /var/log/syslog
```

Use for:

- Find & replace
- Delete lines
- Extract ranges
- Format output



11. LOG FILTERING WORKFLOW



1. Identify Source
contains the data.

Examples:

- auth.log
- syslog
- dmesg
- journalctl
- kern.log

→ 2. Define Time Range



Limit logs to specific timeframe.

```
$ journalctl --since
  "2 hours ago"
$ tail -n 500
  /var/log/syslog
```

→ 3. Filter by Level



Filter by severity or keywords.

```
$ journalctl -p err
$ grep -i error
$ grep -E "fail|denied"
```

→ 4. Search & Extract



Find patterns and extract key fields.

```
$ grep -n "timeout" file
$ awk '{print $1,$5}' file
$ sed -n '50,100p' file
```

→ 5. Correlate Events



Match events across multiple logs.

- auth.log
- syslog
- dmesg
- application logs

→ 6. Draw Findings



Document what you discovered.

- Timeline
- Root cause
- Evidence
- Impact

12. INCIDENT INVESTIGATION CHECKLIST

☐	Define the problem:	What happened? When did it start?
☐	Identify affected systems:	Which servers or services are impacted?
☐	Collect logs:	Gather logs from all relevant sources.
☐	Establish timeline:	Build a timeline of events.
☐	Look for errors:	Search for errors, failures, timeouts.
☐	Check authentication:	Any failed logins or privilege escalation?
☐	Check resource usage:	CPU, Memory, Disk, Network.
☐	Correlate events:	Tie logs together to find the root cause.
☐	Identify root cause:	What is the primary cause of the issue?
☐	Validate & fix:	Apply the fix and monitor.
☐	Document incident:	Write RCA and lessons learned.

13. COMMON LOG LEVELS (PRIORITY)

Level	Priority	Use
emerg	0	System is unusable
alert	1	Action must be taken
crit	2	Critical conditions
err	3	Error conditions
warning	4	Warning conditions
notice	5	Normal but significant
info	6	Informational messages
debug	7	Debug-level messages

14. COMMON LOG LOCATIONS

/var/log/syslog	– System log (Debian/Ubuntu)
/var/log/messages	– System log (RHEL/CentOS)
/var/log/auth.log	– Authentication log
/var/log/kern.log	– Kernel log
/var/log/dmesg	– Kernel ring buffer
/var/log/secure	– Security log (RHEL/CentOS)
/var/log/boot.log	– Boot log
/var/log/nginx/	– Nginx logs
/var/log/audit/audit.log	– Audit log

15. QUICK REFERENCE COMMANDS

```
$ journalctl -xe
$ journalctl -u <service> --since "1 hour ago"
$ tail -f /var/log/syslog
$ grep -i error /var/log/syslog
$ grep -E "fail|denied" /var/log/auth.log
$ awk '{print $1,$5}' /var/log/syslog
$ sed -n '50,100p' /var/log/syslog
$ dmesg -T | less
$ dmesg | grep -i error
$ less /var/log/auth.log
$ tail -F /var/log/<file>.log
```



PRO TIP

Always use timestamps, correlate logs across sources, and keep your system time synchronized using NTP.

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Linux Package Management Cheat Sheet

1. APT (Debian / Ubuntu)

```
# Update package index
$ sudo apt update
# Upgrade packages
$ sudo apt upgrade
# Install package
$ sudo apt install <package>
# Remove package
$ sudo apt remove <package>
# Purge (remove config files)
$ sudo apt purge <package>
# Search package
$ apt search <keyword>
# Show package info
$ apt show <package>
```

Key Files
- /etc/apt/sources.list
- /etc/apt/sources.list.d/
- /var/lib/apt/lists/

2. DPKG (Debian Package Tool)

```
# Install .deb package
$ sudo dpkg -i package.deb
# Remove package
$ sudo dpkg -r <package>
# Purge package
$ sudo dpkg -P <package>
# List installed packages
$ dpkg -l
# Verify package
$ dpkg -V <package>
# Check package status
$ dpkg -s <package>
```

Key Points
- Works with .deb files only
- Does NOT resolve dependencies
- Use apt-get -f install to fix broken dependencies

3. YUM (RHEL/CentOS 7 and older)

```
# Update repos
$ sudo yum makecache
# Install package
$ sudo yum install <package>
# Remove package
$ sudo yum remove <package>
# Search package
$ yum search <keyword>
# List installed
$ yum list installed
# Show package info
$ yum info <package>
# Clean cache
$ sudo yum clean all
```

Key File
- /etc/yum.repos.d/*.repo

4. DNF (RHEL/CentOS 8/9, Fedora)

```
# Update repos
$ sudo dnf makecache
# Install package
$ sudo dnf install <package>
# Remove package
$ sudo dnf remove <package>
# Search package
$ dnf search <keyword>
# List installed
$ dnf list installed
# Show package info
$ dnf info <package>
# Clean cache
$ sudo dnf clean all
```

Key File
- /etc/yum.repos.d/*.repo

5. RPM (Red Hat Package Manager)

```
# Install .rpm package
$ sudo rpm -ivh package.rpm
# Upgrade .rpm package
$ sudo rpm -Uvh package.rpm
# Query package
$ rpm -qi <package>
# List files in package
$ rpm -ql <package>
# Verify package
$ rpm -V <package>
# Remove package
$ sudo rpm -e <package>
```

Key Points:
- Works with .rpm files only
- Does NOT resolve dependencies
- Use with YUM/DNF for dependency handling

6. REPOSITORY WORKFLOW



Best Practices
- Use official repositories when possible.
- Use internal/mirrored repositories for reliability.
- Always update repository metadata before installs.
- Keep repositories minimal and trusted.

7. PACKAGE VERIFICATION

Verify Installed Package

```
# dpkg
$ dpkg -V <package>
# rpm
$ rpm -V <package>
# dnf / yum
$ rpm -Va  (verify all)
```

Verify Package File

```
# Debian (.deb)
$ dpkg-deb --info package.deb
$ dpkg-deb --contents package.deb
# RPM (.rpm)
$ rpm -K package.rpm     # check GPG signature
$ rpm -qip package.rpm   # show info
$ rpm -qlp package.rpm   # list files
```

Tip

Always verify GPG keys for third-party repositories. Import keys and store in /etc/apt/trusted.gpg.d/ or /etc/pki/rpm-gpg/

8. DEPENDENCY MANAGEMENT

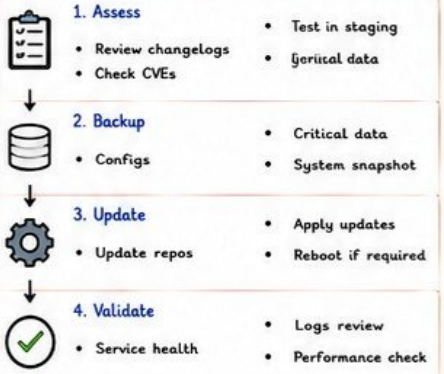
Check Dependencies

```
# apt
$ apt depends <package>
# dnf / yum
$ dnf repoquery --requires <package>
# rpm
$ rpm -qR <package>
```

Fix Broken Dependencies

```
# apt
$ sudo apt --fix-broken install
# dnf
$ sudo dnf install --best --allowerasing
# yum
$ sudo yum install -y --skip-broken
```

9. UPDATE STRATEGY



10. ROLLBACK CONSIDERATIONS

APT (Debian / Ubuntu)

```
$ sudo apt install <package>=<version>
$ sudo apt-mark hold <package>
```

DNF / YUM (RHEL / Fedora)

```
$ sudo dnf history list
$ sudo dnf history undo <ID>
$ sudo dnf downgrade <package>-<version>
```

RPM

```
$ sudo rpm -Uvh --oldpackage <package>.rpm
$ sudo rpm -e <package>
```

Important
- Always test updates in non-production first.
- Keep previous versions of critical packages.
- Back up configurations before major updates.

11. ENTERPRISE RECOMMENDATIONS

Standardize	Automate	Security First	Repository Hygiene	Change Control	Monitoring	Minimal Systems
Use approved repos and versions across environments.	Use Ansible, Puppet, Chef, or SaltStack for patching.	Apply security updates promptly. Monitor CVEs and vendor advisories.	Remove unused repos. Verify repo URLs and GPG keys.	Track changes. Document updates. Maintain audit trails.	Monitor package versions and vulnerabilities.	Install only required packages. Reduce attack surface.

Remember: Good package management = Stable systems, fewer incidents, stronger security.

Pro Tip: Use automation + testing + monitoring to manage packages at scale.

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Linux Performance Monitoring Cheat Sheet

1. top – Process Overview

Real-time view of processes and system resource usage.

```
$ top
```

Key Columns:

- %CPU – CPU usage
- %MEM – Memory usage
- RES – Resident memory
- VIRT – Virtual memory
- PID – Process ID
- TIME+ – CPU time

2. htop – Interactive top

Enhanced top with colors, graphs and easy process management.

```
$ htop
```

Keys:

F1 Help F2 Setup F3 Search
F4 Filter F5 Tree F6 Sort
F9 Kill F10 Quit

Use:

- Easier navigation
- Kill, renice, sort processes

3. vmstat – Virtual Memory Statistics

Report on processes, memory, swap, IO, system and CPU.

```
$ vmstat 1
```

Key Columns:

- r – Running processes
- b – Blocked processes
- swpd – Swap used (KB)
- free – Free memory (KB)
- si/so – Swap in/out (KB/s)
- bi/bo – Block in/out (blocks/s)

4. mpstat – CPU Usage per Core

Report CPU usage for all or individual cores.

```
$ mpstat -P ALL 1
```

Key Columns:

- %usr – User CPU
- %sys – System CPU
- %iowait – IO Wait CPU
- %idl – Idle CPU

5. pidstat – Per Process Statistics

Report CPU, memory, IO for individual processes.

```
$ pidstat -u -r -d 1
```

Options:

-u CPU usage
-r Memory usage
-d IO usage
-p <PID> Specific process
-t Show threads

6. iostat – IO Statistics

Report CPU and device IO stats.

```
$ iostat -xz 1
```

Key Columns (Device):

- r/s – Reads per second
- w/s – Writes per second
- rkB/s – Read KB per second
- wkB/s – Write KB per second
- await – Avg IO wait (ms)
- %util – Device utilization

7. iotop – IO Usage by Process

Display top IO consuming processes.

```
$ iotop
```

Key Columns:

- DISK READ – Read rate
- DISK WRITE – Write rate
- SWAPIN – Swap in rate
- IO% – IO percentage

Use:

- Press P to sort by IO%

8. sar – System Activity Reporter

Collect, report and save system performance data.

```
$ sar -u 1 5
```

Options:

-u CPU usage
-r Memory usage
-d Disk usage
-b I/O and transfer
-q Load average
-n DEV Network stats

Use:

- Historical trends (from sysstat)

9. free – Memory Usage

Display total, used and free memory in the system.

```
$ free -h
```

Key Columns:

- total – Total memory
- used – Used memory
- free – Free memory
- buff/cache – Buffer + Cache
- available – Available memory

10. uptime – System Uptime & Load

Show uptime, users and load averages.

```
$ uptime
```

Shows:

- Current time
- System uptime
- Logged in users
- Load averages (1, 5, 15 min)

11. LOAD AVERAGE INTERPRETATION

Load average shows the average number of processes in runnable (R) or uninterruptible (D) state.

Load vs CPU Cores	Interpretation	Action
Load < Cores	System is healthy	No action needed
Load = Cores	System is fully utilized	Monitor closely
Load > Cores	System is overloaded	Investigate
High iowait	Waiting for disk IO	Check disks
High run queue (r)	CPU contention	Optimize or scale
High memory used	Memory pressure	Free memory / Add RAM

Load Average Example (uptime output)

```
12:30:45 up 25 days,  4:12,  2 users,  load average: 0.45, 0.70, 0.65
                                                     1 min  5 min  15 min
```

- 0.45 (1 min): short-term load
- 0.70 (5 min): medium-term load
- 0.65 (15 min): long-term load

12. QUICK COMMAND REFERENCE

Task	Command
Check overall system	top
Interactive process view	htop
CPU per core	mpstat -P ALL 1
Process CPU/Memory	pidstat -u -r 1
Memory summary	free -h
Virtual memory stats	vmstat 1
Disk IO summary	iostat -xz 1
Disk IO by process	iotop
Historical performance	sar -A
System uptime & load	uptime

13. RESOURCE THRESHOLDS (GUIDE)



CPU

- %util > 80% = High
- %iowait > 20% = IO bottleneck



Memory

- Used > 80% = High
- Available < 10% = Memory pressure
- Swap used > 0 = Investigate



Disk

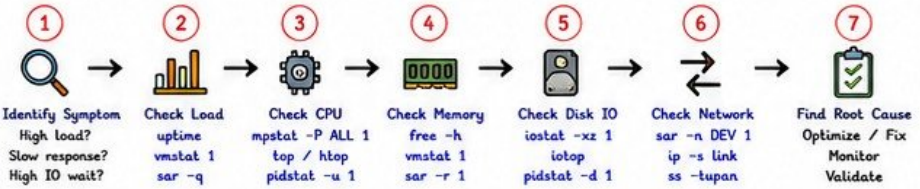
- %util > 80% = Disk busy
- await > 20ms = Slow disk



Load Average

- Load > CPU cores = Overloaded

14. PERFORMANCE INVESTIGATION WORKFLOW



Common Root Causes

- Too many processes
- CPU intensive process
- High disk IO / slow disk
- Memory leak
- Swapping
- Misconfigured services
- Network congestion
- Inefficient queries
- Hardware limits

Tips

- Always correlate metrics.
- Check trends, not just instant values.
- Use sar for historical data.
- Monitor after every change.
- Baseline a healthy system.

15. ENTERPRISE RECOMMENDATIONS

- [x] Collect baseline metrics regularly.
- [x] Use monitoring stack (Prometheus, Grafana, ELK).
- [x] Set alerts on critical thresholds.
- [x] Store historical data (at least 30 days).
- [x] Automate health checks.
- [x] Review performance after deployments.
- [x] Capacity plan based on trends.



REMEMBER:
Healthy systems are not about low usage,
It's about predictable and consistent behavior.

Monitor. Analyze. Optimize.
Build reliable systems.

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Linux Security Cheat Sheet

Secure Systems. Protect Data. Stay Vigilant.

1. SSH HARDENING

- Use key authentication
- Disable root login
- Change default port
- Limit users
- Disable password auth

/etc/ssh/sshd_config

```
Port 2222
PermitRootLogin no
PasswordAuthentication no
AllowUsers admin devops
PubkeyAuthentication yes
MaxAuthTries 3
LoginGraceTime 30
```

Apply changes:

```
$ sudo systemctl reload sshd
```

2. FIREWALL OVERVIEW

UFW (Uncomplicated Firewall)

```
$ sudo ufw status
$ sudo ufw enable
$ sudo ufw default deny incoming
$ sudo ufw default allow outgoing
$ sudo ufw allow 22/tcp
$ sudo ufw allow 80,443/tcp
$ sudo ufw delete allow 22/tcp
```

IPTABLES (legacy)

```
$ iptables -L -n -v
$ iptables -A INPUT -p tcp --dport 22 -j ACCEPT
$ iptables-save > /etc/iptables/rules.v4
```

3. FILE PERMISSIONS

Permission String: -rwxr-xr--

User ↓ Group ↓ Others ↓

Numeric: rwx =7 rx =5 r-x =5 r-- =4 --- =0

```
$ ls -l file.txt
$ chmod 750 script.sh
$ chmod u-x script.sh
$ chown user:group file.txt
$ chgrp group file.txt
```

Special Permissions

- Setuid (4)
- Setgid (2)
- Sticky (1)

```
$ chmod 4755 app
$ chmod 2750 dir
$ chown 1777 /tmp
```

4. SUDO PRACTICES

- Use least privilege
- Grant specific commands
- Avoid NOPASSWD: ALL
- Log administrative actions

Edit sudoers safely:

```
$ sudo visudo
```

Example (least privilege):

```
devops ALL=(root) NOPASSWD:
 /usr/bin/systemctl restart nginx
```

Validate sudoers:

```
$ sudo visudo -c
```

5. PASSWORD POLICY

- Enforce strong passwords
- Set expiration
- Prevent reuse
- Lock accounts after failures

/etc/login.defs

```
PASS_MAX_DAYS   90
PASS_MIN_DAYS    7
PASS_WARN_AGE    7
```

Useful commands

```
$ chage -l username
$ passwd -S username
```

6. SELINUX BASICS

- Mandatory Access Control
- Enforces least privilege
- Context-based security

Check status:

```
$ getenforce
$ sestatus
```

Modes:

Enforcing – Enforce policy
Permissive – Log violations only
Disabled – No SELinux

Set mode:

```
$ sudo setenforce 0    # permissive
$ sudo setenforce 1    # enforcing
```

7. APPARMOR BASICS

- Mandatory Access Control
- Process confinement
- Profile-based security

Check status:

```
$ sudo aa-status
```

Manage profiles:

```
$ sudo aa-enforce /etc/apparmor.d/nginx
$ sudo aa-disable /etc/apparmor.d/nginx
$ sudo aa-logprof          # tune profiles
```

Profile location:

/etc/apparmor.d/

8. FAIL2BAN

- Blocks malicious login attempts
- Protects against brute force
- Monitors logs and bans IPs

Install:

```
$ sudo apt install fail2ban
# or: sudo yum install fail2ban
```

Check status:

```
$ sudo systemctl status fail2ban
$ sudo fail2ban-client status
```

Jail config:

/etc/fail2ban/jail.local

9. AUDIT LOGS

- Track system activity
- Detect suspicious actions
- Ensure compliance

Check audit service:

```
$ sudo systemctl status auditd
```

Search logs:

```
$ sudo ausearch -m USER_LOGIN
$ sudo ausearch -f /etc/passwd
```

View rules:

```
$ sudo auditctl -l
```

10. FILE INTEGRITY

- Detect unauthorized changes
- Monitor critical files
- Early threat detection

Use AIDE:

```
$ sudo apt install aide
$ sudo aideinit
$ sudo aide --check
```

Use Tripwire (Enterprise):

```
$ sudo tripwire --check
```

11. SECURITY CHECKLIST

- [x] SSH key auth enabled
- [x] Root login disabled
- [x] Firewall enabled and configured
- [x] Unnecessary services disabled
- [x] Strong password policy enforced
- [x] System updated regularly
- [x] Logs monitored and reviewed
- [x] File integrity monitoring active
- [x] Backup and recovery tested
- [x] Least privilege access enforced

12. HARDENING REMINDERS

- Keep systems updated
- Remove unnecessary packages
- Disable unused services
- Use strong encryption (TLS)
- Enable MFA where possible
- Monitor logs and alerts
- Follow principle of least privilege
- Document and review configurations
- Test security controls regularly
- Stay informed on vulnerabilities

13. QUICK COMMAND REFERENCE

System Info

```
$ uname -a
$ cat /etc/os-release
$ uptime
$ whoami
$ id
```

Users & Groups

```
$ cut -d: -f1 /etc/passwd
$ groups username
$ sudo usermod -aG sudo user
$ sudo deluser user group
```

Network Security

```
$ ss -tulpen
$ netstat -tulpen
$ sudo ufw status
$ sudo iptables -L
```

Service Security

```
$ systemctl list-units --type=service
$ systemctl is-enabled service
$ systemctl is-active service
$ sudo systemctl disable service
```

File Security

```
$ ls -l /etc
$ find / -perm -4000 -type f 2>/dev/null
$ find / -perm -2000 -type f 2>/dev/null
```

Logs

```
$ journalctl -xe
$ tail -f /var/log/auth.log
$ tail -f /var/log/secure
```

14. INCIDENT RESPONSE QUICK STEPS

1. **Identify** — Confirm the incident.
2. **Analyze** — Collect logs and evidence.
3. **Contain** — Isolate affected systems.
4. **Eradicate** — Remove threat and vulnerabilities.
5. **Recover** — Restore services securely.
6. **Review** — Post-incident review and improve.

PRO TIP

Security is not a one-time task.
It is a continuous process.
Monitor. Harden. Review.
Repeat.

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Linux Bash Scripting Cheat Sheet

Automate Everything. Repeat Nothing.

1. VARIABLES

- No spaces around =
- Case sensitive
- Use $ to access
- Use {} for clarity

```
name="Veriqta"
age=27
echo "Name: $name"
echo "Age: ${age}"
unset name
```

2. FUNCTIONS

- Use function keyword
- Call by name
- Use return for exit code

```
greet() {
  echo "Hello, $1!"
  return 0
}
greet "Engineer"
echo "$?"    # last exit code
```

3. CONDITIONALS

- Use test [ ] or [[ ]]
- String and numeric comparisons

```
if [[ $age -ge 18 ]]; then
  echo "Adult"
elif [[ $age -ge 13 ]]; then
  echo "Teen"
else
  echo "Child"
fi
```

4. LOOPS

- for, while, until
- Loop over lists, ranges or files

```
# for loop
for f in *.log; do
  echo "File: $f"
done
# while loop
while read line; do
  echo "$line"
done < input.txt
```

5. ARRAYS

- Indexed arrays
- Zero-based index

```
fruits=(apple banana cherry)
echo "First: ${fruits[0]}"
echo "All: ${fruits[@]}"
echo "Count: ${#fruits[@]}"
```

6. EXIT CODES

- 0 = Success
- Non-zero = Failure
- $? holds last command's code

```
ls /etc/passwd
echo "Exit code: $?"
if [ $? -ne 0 ]; then
  echo "Command failed!"
fi
```

7. POSITIONAL PARAMETERS

- $0 = script name
- $1, $2 ... arguments
- $# = number of args
- $@ or $* = all args

```
echo "Script: $0"
echo "First arg: $1"
echo "All args: $@"
echo "Count: $#"
# Usage: ./script.sh one two three
```

8. QUOTING

- "double quotes" expand variables
- 'single quotes' literal
- Backslash escapes

```
name="Veriqta"
echo "Hello $name"
echo 'Hello $name'
echo "Path: \$HOME/data"
```

9. ERROR HANDLING

- Use set -e to exit on error
- Check $? for command status
- Use traps for cleanup

```
set -euo pipefail
trap 'echo "Error on line $LINENO" ERR
command_that_might_fail
echo "This runs only if no error"
```

10. DEBUGGING OPTIONS

- -x : trace commands
- -v : verbose
- -n : syntax check only

```
bash -x script.sh    # trace
bash -v script.sh    # verbose
bash -n script.sh    # syntax check
set -x               # in script
set +x               # disable trace
```

11. SCRIPT TEMPLATE

```
#!/bin/bash
# Description: Your script here
set -euo pipefail

# === CONFIG ===
VAR="value"

# === FUNCTIONS ===
function_name() {
   # code
}

# === MAIN ===
main() {
   # main logic
}

main "$@"
exit 0
```

12. BEST PRACTICES

- [x] Always use set -euo pipefail
- [x] Quote your variables
- [x] Validate inputs
- [x] Use meaningful names
- [x] Keep functions small
- [x] Add comments
- [x] Handle errors gracefully
- [x] Log important actions
- [x] Make scripts idempotent
- [x] Test in safe environment

13. COMMON CONDITIONS

Test	Description
-eq	Equal (numbers)
-ne	Not equal
-gt	Greater than
-lt	Less than
-ge	Greater or equal
-le	Less or equal
=	String equal
!=	String not equal
-z	String is empty
-n	String is not empty
-f	File exists
-d	Directory exists
-r	Readable
-w	Writable
-x	Executable

14. COMMON REDIRECTIONS

Operator	Meaning
>	Redirect output (overwrite)
>>	Redirect output (append)
2>	Redirect error (overwrite)
2>>	Redirect error (append)
&>	Redirect both stdout and stderr
<	Input redirection
<<EOF	Here document

Example

```
echo "Hello" > out.txt
echo "Error"  2> err.log
cat file.txt | grep "error" >> report.log
cmd &> all.log
cat <<EOF
This is
a here document
EOF
```

15. USEFUL BUILT-IN COMMANDS

echo	Print text
read	Read user input
test / [ ]	Evaluate conditions
expr	Evaluate expressions
printf	Formatted output
local	Local variable (in function)
export	Export variable
trap	Trap signals
shift	Shift positional parameters
return	Return from function

16. QUICK SCRIPTING WORKFLOW

1. **Plan** — Define the task and logic
2. **Write** — Create script step by step
3. **Test** — Test with sample inputs
4. **Debug** — Fix issues, add logs
5. **Harden** — Add error handling, validation
6. **Deploy** — Run in production safely
7. **Monitor** — Check logs and performance

PRO TIP

Small scripts save hours.
Consistent scripts save careers.

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Linux Troubleshooting Cheat Sheet

Diagnose Fast. Investigate Deep. Fix Right.

1. CPU ISSUES

Symptoms
- High CPU usage
- Load average high
- System slow / unresponsive

Investigate

```
$ top
$ htop
$ ps -eo pid,ppid,cmd,%cpu --sort=-%cpu
  | head
$ mpstat -P ALL 1 5
$ vmstat 1 5
```

Fix Ideas
- Kill runaway process
- Optimize application
- Check cron jobs
- Check for CPU intensive scripts

2. MEMORY ISSUES

Symptoms
- High memory usage
- OOM kills
- System swapping

Investigate

```
$ free -h
$ top
$ vmstat 1 5
$ cat /proc/meminfo
$ dmesg | grep -i oom
```

Fix Ideas
- Identify memory hungry process
- Increase RAM / Optimize app
- Disable unnecessary services
- Add swap if required

3. DISK ISSUES

Symptoms
- Disk full
- High I/O wait
- Slow read/write

Investigate

```
$ df -h
$ du -sh /* 2>/dev/null | sort -h
$ iostat -xz 1 5
$ iotop -oPa
$ dmesg | grep -i 'I/O error'
```

Fix Ideas
- Free up disk space
- Rotate / Clean logs
- Fix filesystem errors (fsck)
- Check disk health (smartctl -a /dev/sdX)

4. NETWORK ISSUES

Symptoms
- Cannot connect
- Slow network
- High packet loss

Investigate

```
$ ip a
$ ip r
$ ping <host>
$ traceroute <host>
$ ss -tulpen
$ ethtool <iface>
$ tcpdump -i <iface>
```

Fix Ideas
- Check IP / Gateway / DNS
- Restart network service
- Check firewall rules
- Check cable / Interface status

5. SERVICE FAILURES

Symptoms
- Service not starting
- Service crashes
- Dependency issues

Investigate

```
$ systemctl status <service>
$ systemctl --failed
$ journalctl -u <service> --no-pager -n 100
$ systemctl cat <service>
```

Fix Ideas
- Check logs for errors
- Fix configuration
- Check dependencies
- Restart / Reload / Reinstall

6. PERMISSION ERRORS

Symptoms
- Permission denied
- Cannot access files/dirs
- Operation not permitted

Investigate

```
$ ls -l <file/dir>
$ namei -l <path>
$ id
$ getfacl <file>
$ ausearch -m avc -ts recent
```

Fix Ideas
- Fix file/dir ownership (chown)
- Set correct permissions (chmod)
- Check ACLs
- Review sudoers if needed

7. DNS FAILURES

Symptoms
- Domains not resolving
- Intermittent resolution
- Slow DNS response

Investigate

```
$ resolvectl status
$ dig <domain>
$ nslookup <domain>
$ cat /etc/resolv.conf
$ systemd-resolve --statistics
```

Fix Ideas
- Check DNS server
- Flush cache (resolvectl flush-caches)
- Update resolv.conf / NetworkManager
- Check firewall / Network connectivity

8. LOG INVESTIGATION

Common Logs
- /var/log/syslog
- /var/log/messages
- /var/log/auth.log
- /var/log/kern.log
- /var/log/dmesg

Commands

```
$ journalctl -xe
$ tail -f /var/log/syslog
$ grep -i "error\|fail\|warn" /var/log/syslog
$ dmesg -T | less
```

Tips
- Check timestamps around the issue
- Search for ERROR / FAIL / WARN
- Correlate logs across services

9. PROCESS INVESTIGATION

Goals
- Find rogue processes
- Understand process tree
- Find resource usage

Commands

```
$ ps aux --sort=-%cpu | head
$ pstree -p <pid>
$ lsof -p <pid>
$ cat /proc/<pid>/status
$ strace -p <pid> -f -tt
```

Tips
- Identify parent/child processes
- Check open files and connections
- Use strace for deep debugging

10. RECOVERY WORKFLOW

1. **Detect** — Alert / User reports issue
2. **Triage** — Confirm issue, Identify impact
3. **Investigate** — Collect evidence, Check metrics, logs, configs
4. **Identify Cause** — Find root cause(s)
5. **Fix** — Implement solution
6. **Verify** — Confirm system is healthy
7. **Document** — Post-incident notes & learnings

Key Principle:
Follow the process. Don't guess. Collect data, form a hypothesis, test it, and verify the result.

11. ROOT CAUSE ANALYSIS CHECKLIST

- ☑ What is the exact problem?
- ☑ When did it start? Any recent changes?
- ☑ What is the impact? Who is affected?
- ☑ What symptoms are visible?
- ☑ What do the logs show?
- ☑ What metrics confirm the issue?
- ☑ What changed recently (deploy, config, patch)?
- ☑ Is this reproducible?
- ☑ What triggered the issue?
- ☑ What is the underlying cause (not the symptom)?
- ☑ What is the permanent fix?
- ☑ How can we prevent this in future?
- ☑ What monitoring/alerts can detect earlier?
- ☑ Document findings and actions.

Common Tools Quick List

top, htop	– CPU / Processes
free, vmstat	– Memory
df, du, iostat	– Disk
ip, ping, ss	– Network
systemctl, journalctl	– Services
dig, nslookup	– DNS
grep, tail, less	– Logs

PRO TIP
- Monitor before users complain.
- Automate checks.
- Keep systems updated.
- Review logs daily.
- Document everything.
- Small issues ignored become big outages.

HARDENING REMINDERS
- ✔ Keep OS & packages updated
- ✔ Use least privilege
- ✔ Disable unused services
- ✔ Use strong passwords / SSH keys
- ✔ Enable firewall
- ✔ Monitor & review logs
- ✔ Backup regularly
- ✔ Test recovery plans

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Linux Command Reference Master Cheat Sheet

One Page. Every Command. Everyday Power.

1. ESSENTIAL FILE COMMANDS

ls -lah	List files (long format)
cd /path	Change directory
pwd	Print working directory
mkdir dir	Create directory
rmdir dir	Remove empty directory
cp src dest	Copy file or directory
mv src dest	Move or rename
rm file	Remove file
rm -rf dir	Remove directory recursively
touch file	Create empty file
cat file	Display file content
less file	View file with pager
head file	Show first 10 lines
tail -f file	Follow file content

2. PROCESS COMMANDS

ps aux	Show all processes
top	Real-time process view
htop	Interactive process viewer
pgrep name	Find process by name
pidof name	Get PID of process
kill PID	Terminate process
kill -9 PID	Force kill process
nice -n 10 cmd	Run with priority
renice -n 5 -p PID	Change priority
bg	Resume in background
fg	Foreground process
jobs	List background jobs

3. NETWORKING COMMANDS

ip a	Show IP addresses
ip r	Show routing table
ping host	Test connectivity
traceroute host	Trace network path
ss -tulnp	Show listening ports
netstat -tulnp	(legacy)
curl -I url	Check HTTP headers
wget file	Download file
dig domain	DNS lookup
nslookup domain	DNS query
hostname -I	Show IP address
nmap -sT host	Port scan

4. STORAGE COMMANDS

df -h	Disk usage
du -sh dir	Directory size
lsblk	List block devices
fdisk -l	List partitions
blkid	Show block UUIDs
mount /dev/sdX /mnt	Mount device
umount /mnt	Unmount device
sync	Flush disk buffers
hdparm -Tt /dev/sdX	Disk speed test

5. USER MANAGEMENT

whoami	Current user
id	Show user and groups
users	Logged-in users
adduser user	Add new user
usermod -aG sudo user	Add user to group
passwd user	Set user password
deluser user	Delete user
groupadd devops	Create group
groupdel devops	Delete group
groups user	Show user groups

6. PERMISSION COMMANDS

chmod 755 file	Change file mode
chmod -R 755 dir	Recursive chmod
chown user file	Change owner
chown -R user:group dir	Recursive owner
chgrp group file	Change group
umask 022	Set default permissions
ls -l	Show permissions
getfacl file	View ACL
setfacl -m u:user:rwx file	Set ACL

7. SERVICE COMMANDS

systemctl start svc	Start service
systemctl stop svc	Stop service
systemctl restart svc	Restart service
systemctl reload svc	Reload service
systemctl status svc	Service status
systemctl enable svc	Enable on boot
systemctl disable svc	Disable on boot
journalctl -u svc -f	View service logs

8. PACKAGE COMMANDS

# Debian/Ubuntu (APT)	
sudo apt update	Update index
sudo apt upgrade	Upgrade packages
sudo apt install pkg	Install package
sudo apt remove pkg	Remove package
sudo apt autoremove	Clean unused
# RHEL/CentOS/Fedora (YUM/DNF)	
sudo dnf check-update	Check updates
sudo dnf install pkg	Install package
sudo dnf remove pkg	Remove package
sudo dnf clean all	Clean cache

9. MONITORING COMMANDS

uptime	System uptime
free -h	Memory usage
vmstat 1	System stats
mpstat -P ALL 1	CPU usage per core
iostat -xz 1	Disk I/O stats
df -h	Disk usage
ss -tulnp	Network sockets
watch -n 2 cmd	Repeat command
sar -u 1 5	CPU report
sar -r 1 5	Memory report

10. COMPRESSION COMMANDS

tar -czf file.tar.gz dir	Create tar.gz
tar -xzf file.tar.gz	Extract tar.gz
tar -cjf file.tar.bz2 dir	Create tar.bz2
tar -xjf file.tar.bz2	Extract tar.bz2
gzip file	Compress file
gunzip file.gz	Decompress file.gz
zip -r file.zip dir	Create zip
unzip file.zip	Extract zip
xz file	Compress with xz
unxz file.xz	Decompress xz

11. SEARCH COMMANDS

find /path -name file	Find file by name
find /type f -size +100M	Find by size
grep 'text' file	Search text
grep -R 'text' /path	Recursive grep
locate filename	Fast file search
which cmd	Locate command
whereis cmd	Locate binary/source
type cmd	Show command type

12. TEXT PROCESSING COMMANDS

cat file	Display content
less file	View file
head -n 20 file	First 20 lines
tail -n 20 file	Last 20 lines
cut -d',' -f1 file	Cut columns
sort file	Sort lines
uniq file	Remove duplicates
wc -l file	Count lines
awk '{print $1}' file	Process fields
sed 's/old/new/g' file	Replace text

13. SCHEDULING COMMANDS

crontab -e	Edit cron jobs
crontab -l	List cron jobs
crontab -r	Remove cron jobs
# Example cron (every day 2:30 AM)	
30 2 * * * /path/script.sh	
at 10:00	Schedule one-time job
atq	List at jobs
atrm <jobid>	Remove at job

14. LOGGING COMMANDS

journalctl	View all logs
journalctl -xe	View logs (detailed)
journalctl -f	Follow logs live
dmesg | less	Kernel ring buffer
tail -f /var/log/syslog	Syslog file
tail -f /var/log/auth.log	Auth logs
grep 'ERROR' /var/log/syslog	Search errors
logrotate -d /etc/logrotate.conf	Test logrotate

15. LINUX ENGINEER DAILY WORKFLOW

1. CHECK — Uptime, Load, Disk, Memory Health
2. INVESTIGATE — Processes, Logs, Services, Network, Resources
3. RESOLVE — Fix Issues, Restart Services, Optimize
4. VERIFY — Confirm System Health & Stability
5. DOCUMENT — Record Findings, Root Cause, Actions Taken

16. FINAL VERIQTA LINUX MASTERY ROADMAP

FOUNDATION	COMMAND MASTERY	SYSTEM OPERATIONS	NETWORKING	STORAGE MANAGEMENT	SECURITY & HARDENING	AUTOMATION	EXPERT ENGINEER
• Linux Basics	• Navigation	• Processes	• IP & Routing	• Partitions	• Permissions	• Bash Scripting	• Troubleshooting
• File System	• File Management	• Services	• DNS	• File Systems	• SSH Hardening	• Cron Jobs	• Performance Tuning
• Permissions	• Text Processing	• Scheduling	• Ports & Sockets	• Mounting	• Firewall	• Backups	• System Design
• Users & Groups	• Search & Filters	• Logging	• Firewall	• Disk Usage	• Logs & Auditing	• Monitoring	• Reliability
• Essential Commands	• Productivity Boost	• Monitoring	• Troubleshooting	• RAID & LVM Basics	• Best Practices	• Incident Response	• Leadership

CONSISTENCY + PRACTICE + REAL-WORLD EXPERIENCE = LINUX MASTERY

VERIQTA PRINCIPLE

Know the system.
Understand the logs.
Respect the process.
Automate the repeatable.
Document everything.
That's how experts win.

GOLDEN REMINDERS

- ✔ Always backup before changes
- ✔ Test in non-production first
- ✔ Keep systems updated
- ✔ Use least privilege
- ✔ Monitor, Alert, Act
- ✔ Automate the boring tasks
- ✔ Document your environment
- ✔ Never stop learning

YOU ARE BECOMING A LINUX EXPERT!